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COMPUTER AND INFORMATION TECHNOLOGY DEPARTMENT

Web-Based Lost and Found Management System for ANU Students to Improve Recovery Efficiency

By

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Submitted in partial fulfillment of the requirements of the Bachelor of Science in Business and Information Technology

Declaration

The Research Proposal as presented in this report is my original work and has not been presented for any other University Award. Materials of work done by other researchers are mentioned by clear reference.

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The Research Proposal has been submitted in partial fulfillment of the Requirements for the award of a Bachelor of Science in Business and Information Technology at Africa Nazarene University with my approval as the University supervisor.

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Dedication

This project is dedicated to my family and loved ones, whose encouragement, prayers, and unwavering support have continually strengthened me throughout my academic journey. Their belief in my potential has been the foundation on which this work stands.

Acknowledgment

I express my sincere gratitude to the Almighty for granting me strength, good health, and guidance throughout this research. My appreciation also goes to my supervisor, Mr. Joram Makasa, for the constructive feedback, mentorship, and invaluable academic support that shaped this project. I also thank the faculty and staff of the Department of Computer and Information Technology at Africa Nazarene University for providing the resources and environment that made this study possible. Special thanks to my classmates and friends for their encouragement, as well as the participants who offered insights necessary for system requirements gathering.

Abstract

This study presents the development of a Web-Based Lost and Found Management System for Africa Nazarene University (ANU). The system addresses the recurring challenge of students losing personal items on campus and the inefficiencies associated with the manual recovery process. The application integrates core functionalities such as reporting lost items, reporting found items, automated item matching, security officer verification, notifications, and administrative report generation. A descriptive research design guided the study, supported by interviews, questionnaires, and analysis of comparable systems to establish functional and non-functional requirements.

The prototype was developed using Figma, following a user-centered design approach and ANU's visual identity standards. The system's workflow aligns with the institution's operational structure, incorporating students, security officers, and administrators as key user groups. The solution enhances accuracy in matching lost and found items, improves communication through real-time notifications, and provides administrators with structured reports for decision-making.

The results demonstrate that the prototype successfully meets the outlined objectives and offers a viable pathway to digitizing campus lost-and-found operations. Recommendations for future development include implementing a fully functional web-based application, integrating NFC or QR-code tagging for frequently lost items, and deploying a centralized cloud-based database for real-time synchronisation.

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List of Abbreviations

This section presents the abbreviations and acronyms used throughout the project for clarity and consistency.

Abbreviation	Full Meaning
ANU	Africa Nazarene University
KAA	Kenya Airports Authority
NTSA	National Transport and Safety Authority
UoN	University of Nairobi
SAPS	South African Police Service
ACSA	Airports Company South Africa
API	Application Programming Interface
ID	Identification Document
ERD	Entity Relationship Diagram
DFD	Data Flow Diagram
UI	User Interface
UAT	User Acceptance Testing
SQL	Structured Query Language
MySQL	My Structured Query Language
ICT	Information and Communication Technology
AI	Artificial Intelligence
PDF	Portable Document Format
UI	User Interface
UX	User Experience
API	Application Programming Interface
DBMS	Database Management System
SMS	Short Message Service
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

The management of lost and found items within public institutions is an area that has traditionally relied on manual processes, informal reporting, and physical storage points. Universities, schools, airports, and government facilities have historically used physical notice boards, logbooks, and verbal communication to help individuals recover misplaced belongings. The concept of lost and found management has existed for decades, originating from early public administration practices where custodians or security officers kept handwritten logs of items reported or recovered. These traditional approaches were adequate when populations were smaller and mobility was limited, but as institutions expanded and foot traffic increased, the limitations of manual lost and found processes became increasingly evident (Smith, 2019). In university environments, where students frequently move across multiple buildings while carrying essential personal items, the probability of misplacement becomes higher, making lost and found systems a necessary part of campus service delivery.

In recent years, technological advancement has transformed the way institutions organize their internal services, with digital systems now being integrated into libraries, health services, attendance tracking, and academic administration. Web technology, in particular, has introduced opportunities for real-time reporting, digital record-keeping, and accessible service platforms within university settings. Studies show that web applications enhance service efficiency, reduce communication barriers, and enable faster information retrieval (Adams & Njoroge, 2020). As a result, digital platforms are increasingly being adopted to replace manual lost and found processes. However, while various organizations globally have embraced online reporting tools and web-based tracking systems, similar adoption in Kenyan higher learning institutions remains limited, leaving a significant gap in service efficiency. Most local universities still rely on physical security offices or notice boards as the primary method of reporting lost items, resulting in delays and low recovery rates.

Current contextual factors also reveal the importance of digitizing this area. Modern universities operate in environments where students depend heavily on mobile devices, expect convenient

digital solutions, and require timely communication. Personal items such as laptops, mobile phones, chargers, and student IDs are essential to academic activities, and their loss can significantly disrupt daily routines. Furthermore, research shows that centralized digital systems improve traceability, create structured item records, and reduce the risk of misplaced or unidentified items within institutional environments (Kim & Lee, 2021). With increased student populations and higher dependence on digital tools, institutions like Africa Nazarene University face the need to adopt modern, technology-driven approaches to improve service delivery. The gap between traditional manual reporting and emerging digital systems highlights the importance of developing a Web-Based lost and found platform tailored specifically for campus environments.

1.2 Problem Statement

Students at Africa Nazarene University frequently misplace or lose items such as student identification cards, books, mobile phones, and electronic accessories while moving across various parts of the campus. The current process for recovering these items is largely manual and fragmented, relying on physical notice boards, informal student communication, or security office logs. Research indicates that manual systems lack efficiency, accuracy, and a reliable way to track reported items (Doe, 2021). Although various institutions globally have adopted digital lost and found systems, existing research shows limited application of such technology within Kenyan universities (Omondi & Kariuki, 2022). This reveals a research gap: the lack of a centralized, Web-Based system that facilitates structured reporting, automated matching, and real-time notifications within a university setting. Addressing this gap is crucial because current literature does not offer a well-established solution that fits the needs, scale, and context of ANU's student population.

1.3 Justification of the Problem

The identified problem is significant because the loss of essential personal items can interrupt academic activities, cause distress, and lead to financial burdens for students. A manual lost and found process is not only slow but also unreliable, resulting in low recovery rates and lack of accountability. A digital solution is therefore necessary to streamline the reporting, matching, and verification of lost and found items. Developing a Web-Based system will enhance

efficiency, improve user experience, and provide administrators with accurate data on item recovery trends. By addressing the research gap and modernizing the process, the project contributes to improving student welfare and institutional service delivery.

1.4 Goal of the Study

The goal of this study is to develop a Web-Based Lost and Found Management System for Africa Nazarene University that digitizes the reporting, tracking, matching, and recovery of lost and found items within the campus. The system aims to enhance recovery efficiency, improve communication, support verification processes, and provide administrators with structured data for informed decision-making.

1.5 Objectives of the Study

Main Objective

To design and develop a Web Lost and Found Management System for Africa Nazarene University that streamlines the reporting, verification, matching, and retrieval of lost items.

Specific Objectives

1. To enable students to create and manage personal accounts for reporting and tracking lost or found items within the university.

2. To develop a module that allows users to report lost or found items by capturing details such as item category, description, photos, and location.

3. To implement an admin/security verification system for reviewing, approving, and publishing reported items to ensure accuracy and prevent false reports.

4. To design an automated matching engine that compares lost and found item reports using keywords and categories, and alerts users when a potential match is identified.

5. To provide real-time notifications to students regarding item matches, verification updates, and collection instructions.

6. To generate administrative reports showing matched items, recovery rates, frequently lost items, and system activity logs to support decision-making.

1.6 Research Questions

1. How can a Web-Based system improve the reporting and tracking of lost items among students at Africa Nazarene University?
2. What mechanisms are required to ensure accurate matching between lost and found item reports?
3. How can real-time notifications improve communication and reduce recovery time for lost items?
4. What verification procedures will ensure that recovered items are returned to the rightful owners?
5. How can the system support administrators in monitoring lost and found trends and generating summary reports?

1.7 Limitations of the Study

This study is limited by its scope, as it focuses solely on items lost within Africa Nazarene University and does not integrate external systems such as national ID recovery databases or police registries. Methodologically, the research relies heavily on stakeholder inputs, which may be constrained by availability and access. In terms of resources, the project must be completed within the academic semester, limiting the inclusion of advanced features such as artificial intelligence-based image recognition. Additionally, findings may not be fully generalizable to other universities due to differences in infrastructure, population size, and institutional policies.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to digital lost and found management systems, focusing on local, regional, and international contexts. It examines how institutions in Kenya, across Africa, and globally have implemented systems and technologies designed to support item reporting, recovery, and verification. The review assesses the characteristics, strengths, and shortcomings of these systems, with emphasis on web-based solutions, database structures, notification mechanisms, and administrative workflows. By comparing different implementations, this chapter identifies trends, gaps, and areas where the proposed Lost and Found Management System for Africa Nazarene University provides unique value.

2.2 Local Literature Review (Kenya)

Kenya has made considerable progress in digitizing public services and institutional processes, yet the development of dedicated lost and found management systems remains relatively limited. Most initiatives have been implemented within government agencies, transportation hubs, and financial service providers, rather than within institutions of higher education. The review of local systems provides insight into the technological landscape, capabilities, and existing gaps that influence the design of a web-based lost and found system for Africa Nazarene University.

One of the earliest and most structured lost and found systems in Kenya is found within the **Kenya Airports Authority (KAA)**, which manages property recovered at Jomo Kenyatta International Airport, Moi International Airport, Kisumu International Airport, and various regional airstrips. KAA maintains a digital register of found items accessible to staff through an internal platform (KAA, 2020). The system captures item descriptions, storage locations, timestamps, and claim history. However, passengers do not have direct access to this internal system; instead, they must fill a public inquiry form on the KAA website or contact airport customer care. This approach limits transparency and depends heavily on manual searching by staff members. The lack of a web application and real-time notifications restricts user convenience, highlighting a gap in interactive digital engagement.

Another relevant system is the **National Transport and Safety Authority (NTSA)** lost documents portal, which digitizes the process of checking recovered driver's licenses and identification documents (NTSA, 2021). The online platform displays lists of found documents and includes a search feature for specific names or ID numbers. Although it demonstrates the potential of searchable databases in improving retrieval efficiency, the process remains largely manual. Updates to the platform are not immediate, and users do not receive automated alerts when their documents appear in the registry. Additionally, the service is highly specialized—focused solely on driver's licenses—and does not cover general personal items.

In the education sector, attempts at digitized lost and found systems are minimal. For instance, **Kenyatta University (KU)** operates a basic digital notice board through which students can view posts about found items, predominantly student identification cards (Kenyatta University, 2020). Reports are made manually by administrative offices, and updates are sporadic. The system lacks essential functionalities such as image uploads, categorization of items, student reporting, or matching algorithms. As a result, student recovery rates depend heavily on physical visits to campus security offices.

Similarly, the **University of Nairobi (UoN)** provides a limited web-based notice system for lost student IDs and academic documents. Like KU, the system does not allow users to submit lost item reports online, nor does it incorporate mobile-friendly interfaces or automated alerts. It serves primarily as a digital pin board rather than a full-service lost and found system. These university-level approaches demonstrate that while digital attempts exist locally, they are far from comprehensive and do not reflect the mobility patterns or digital expectations of modern students.

In addition to institutional systems, Kenya has several private-sector innovations that offer relevant insights, even though they do not directly address lost and found management. Tools such as **mSurvey**, **Ona Data**, and **Tanda** have established user-driven reporting, mobile data collection, and dashboard analytics (Ona, 2022). These platforms illustrate the viability of mobile-first solutions and demonstrate Kenya's strong app development capacity. However, none are designed specifically for managing lost and found workflows, leaving a gap for context-specific solutions within educational settings.

Safaricom's **M-PESA ecosystem** and the **Daraja API** also provide foundational technological infrastructure that could support lost and found applications. These tools enable real-time data communication between mobile apps and server-side systems (Safaricom, 2022). Although not intended for lost and found services, they underscore Kenya's readiness for integrated mobile platforms capable of managing dynamic reporting activities.

Another example is the **eCitizen** portal, which supports lost identity document checks across various government databases. Though efficient, the service is limited to identification documents and depends on existing government registries. Again, the focus is on document retrieval rather than comprehensive item management.

Overall, the characteristics of local systems include basic web interfaces, manual update workflows, limited automation, and minimal mobile interaction. The shortcomings across all Kenyan systems—whether institutional, governmental, or private—include lack of automated matching, absence of mobile interfaces, insufficient user engagement, slow update cycles, and dependence on manual workflows. None of the reviewed systems demonstrate full integration of user reporting, administrative verification, photo uploads, category-based matching, and automated notifications in a university environment.

In conclusion, local literature demonstrates the existence of partially digitized processes for item recovery but reveals significant gaps in automation, mobile accessibility, and tailored university workflows. Kenya's institutional systems remain basic and function more as static online notice boards rather than dynamic, user-friendly lost and found platforms. This highlights the need for a web-based, fully interactive system that allows students and staff to seamlessly report, track, match, and recover lost items within a university environment.

2.3 Regional Literature Review (Africa)

Across Africa, digital transformation has accelerated in recent years, with governments and institutions increasingly adopting digital platforms to streamline services. However, lost and found management systems remain underdeveloped across the continent, especially within educational institutions. The regional review focuses on solutions implemented in public

services, airports, law enforcement, and private sectors, highlighting the characteristics and limitations of these systems across Africa.

One of the most structured regional systems is the **South African Police Service (SAPS) Found Property System**, which records items recovered by police and makes them available to the public upon request (SAPS, 2020). The system uses a centralized national database that categorizes found items and logs the circumstances of recovery. Citizens can check with local police stations, but access to databases is largely internal. While SAPS demonstrates the potential for centralized record-keeping, it remains limited by manual verification procedures and lacks features such as mobile integration, user reporting, or automated matching.

Another regional example is the **Airports Company South Africa (ACSA)** lost property platform, which supports international and domestic airports such as OR Tambo International Airport and Cape Town International Airport (ACSA, 2021). ACSA allows passengers to submit inquiries online, listing lost items through structured forms. Airport staff then manually compare reported items with their internal registers. Although functional, the system is heavily dependent on staff intervention and lacks automation features. The platform does not include photo uploads for items found, nor does it support mobile usage beyond a responsive website.

In East Africa, Rwanda's **IremboGov** platform has digitized several public services, including recovery checks for lost national identity cards (IremboGov, 2022). The system is highly efficient and reflects Rwanda's commitment to e-governance. However, it is narrowly focused on identity documents and does not extend to general lost and found items. It also lacks automated item matching, mobile app integration, and institutional workflows needed for campus environments.

Uganda and Tanzania have experimented with semi-digital lost item registries, mostly through police and government portals. However, these systems remain basic, often listing found documents in PDF formats or static web pages (Uganda Police Force, 2020). The listings are updated irregularly and require users to manually scan lists to find their items. This approach does not align with the expectations of modern mobile-first users and fails to provide comprehensive item tracking mechanisms.

Nigeria has seen innovation through private-sector solutions such as **FoundMe Nigeria**, a QR-based tagging system that allows users to attach scannable codes to personal items (FoundMe, 2021). When an item is found, scanning the QR code alerts the owner. While innovative, this solution depends entirely on pre-tagging items and requires the finder to have technical awareness. It does not help in cases where items were not tagged beforehand, nor does it include an administrative backend suitable for institutional environments.

Ethiopia's government has explored document recovery platforms similar to Kenya's eCitizen, but these platforms remain document-focused and not general-purpose lost and found systems. They also rely heavily on manual updates and do not support real-time notifications or algorithmic matching.

When examining African universities specifically, there is limited evidence of structured lost and found management systems. Studies on digital service adoption in universities in Ghana, Uganda, and South Africa indicate that institutions largely focus on academic management systems, library automation, and financial operations (Abdulai, 2020; Mwesigwa, 2021). Services related to student welfare, such as lost and found management, are often overlooked. Most universities continue to use physical notice boards or rely on campus security, leading to inefficiencies in communication and item recovery.

A common characteristic of regional systems is their reliance on central administrative bodies—police departments, airport authorities, or government agencies—to manage lost property records. These systems generally lack user-initiated features, such as direct reporting through mobile apps, photo uploads, or peer-to-peer communication. Additionally, most platforms are not optimized for mobile use, despite Africa's high mobile penetration rates. The absence of automated matching mechanisms or intelligent filtering further reduces the effectiveness of these systems.

Despite these shortcomings, regional technologies and digitization initiatives demonstrate that Africa is ready for web-based solutions. The rise of mobile money, digital ID systems, and online service platforms indicates that citizens are comfortable interacting with digital services

through their phones. This context supports the viability of a web-based lost and found system specifically tailored for university environments.

In summary, while Africa has made progress in digital governance and service delivery, lost and found management systems remain largely manual, fragmented, and underdeveloped. Existing solutions are often limited to specific sectors and do not incorporate modern features such as automation, mobile notifications, or student-oriented workflows. These gaps highlight the opportunity for an innovative, web-first lost and found system that addresses the unique needs of university communities.

2.4 International Literature Review (Global)

Globally, lost and found management systems are more advanced, particularly in regions where digital service delivery is highly developed. Many global solutions incorporate automation, cloud-based infrastructure, and mobile integration. The international review examines systems used in airports, universities, hospitality industries, and public transportation networks, emphasizing their features and limitations.

One of the most sophisticated global platforms is **Chargerback**, a cloud-based lost and found management system widely used across the United States by airlines such as Alaska Airlines, hotels, amusement parks, and other high-traffic venues (Chargerback, 2021). Chargerback incorporates automated matching algorithms, barcode tracking, and courier service integration. Users who lose items can submit detailed reports through a mobile-friendly interface, while staff members can upload photos of found items. The system attempts to match reports using keyword analysis and AI-driven enhancements. Despite its technological strengths, Chargerback operates on a subscription basis, making it financially inaccessible to many educational institutions in developing countries. Moreover, its workflows are designed for commercial environments rather than university campuses.

Another prominent system is **Crowdfind**, which serves airports, museums, universities, and event organizers (Crowdfind, 2021). It introduces modern features such as image recognition and AI-assisted searching. Crowdfind also uses digital claim tickets, automated notifications, and centralized staff dashboards. However, the platform requires high-speed internet access and

subscription payments, limiting its applicability in developing regions. Its university clients often use simplified versions of the system that do not fully exploit automation or mobile integration.

Public transportation systems in Europe and North America also utilize advanced lost property portals. For example, **Transport for London (TfL)** operates a comprehensive digital lost property system that manages thousands of items daily (TfL, 2020). The system includes automated email notifications, online reporting tools, digital claim processing, and integration with inventory management software. TfL's platform is highly efficient but geographically specific and not designed for educational institutions.

Asian countries showcase some of the most technologically advanced systems. Japan's **Lost360** and Korea's **SafeReturn Platform** integrate RFID technology, QR tagging, cloud infrastructure, and AI-driven matching (Lee & Kobayashi, 2022). These systems support city-level coordination between municipal authorities, police departments, and transportation providers. However, these advanced technologies require expensive hardware and maintenance, making them unrealistic for institutions in developing countries, especially universities.

International universities have also implemented digital lost and found systems. Universities such as the **University of California, Stanford University, and the University of Toronto** operate online portals where students can submit lost item reports and view listed found items. These systems typically include photo uploads, item categorization, and administrative dashboards. However, they seldom incorporate automated item matching, AI features, or mobile applications. Most rely on manual verification by campus staff, similar to systems in Africa and Kenya, but with better digital interfaces.

Global systems demonstrate a consistent trend toward innovation, automation, and user-centered design. However, they also reveal limitations when considered in the context of a Kenyan university. Many systems assume reliable high-speed internet, consistent funding, and advanced institutional infrastructure. They also prioritize commercial or public environments rather than student populations. Furthermore, subscription costs and reliance on proprietary technologies make them impractical for institutions like Africa Nazarene University.

In summary, international systems offer valuable insights into the potential of digital lost and found management, showing how automation, cloud computing, mobile integration, and AI can enhance recovery processes. Yet these systems also highlight financial, infrastructural, and contextual barriers that prevent direct adoption in a Kenyan university setting. This underscores the need for a localized, accessible, mobile-first solution tailored to student workflows and campus environments.

2.5 Conclusion

The review of local, regional, and international literature demonstrates varying levels of sophistication in lost and found management systems. Local systems in Kenya are functional but basic, relying heavily on manual processes and lacking mobile integration, automation, or real-time notifications. They illustrate an ongoing transition toward digitization but fall short of supporting dynamic, student-centered reporting and recovery workflows. Regional systems across Africa show some progress, particularly in government-led digital platforms and police property systems, but largely remain limited by technical constraints, manual verification processes, and lack of institutional customization. International systems, meanwhile, display advanced technological capabilities such as AI-driven matching, automated workflows, and cloud-based architectures, but are often costly, infrastructure-dependent, and designed for commercial rather than educational contexts.

Across all levels, a common pattern emerges: while digital lost and found systems exist, they are often sector-specific, non-automated, or limited in accessibility. More importantly, there is a clear absence of web-based, fully integrated systems tailored to university environments in Kenya and much of Africa. Universities experience high item turnover, with students frequently misplacing items such as ID cards, electronics, books, and personal belongings. Yet no reviewed system—local, regional, or global—offers a solution that captures real-time student reporting, automated matching capabilities, administrative verification, mobile notifications, and campus-specific workflows in a unified platform.

The existing literature reveals multiple research gaps:

1. Lack of university-focused lost and found digital systems.

2. Limited mobile-first platforms in Kenya and Africa.
3. Inadequate automation for item matching and verification.
4. Heavy reliance on manual processes.
5. Absence of systems tailored for low-resource, low-cost institutional environments.

The proposed Web-Based Lost and Found Management System for Africa Nazarene University directly addresses these gaps by introducing an accessible, student-centered, mobile platform that enables reporting, searching, matching, and verification of lost and found items. The system's design aligns with contextual needs identified in the literature and provides a more efficient, modern alternative to existing fragmented and manual processes.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology adopted for the development of the Web-Based Lost and Found Management System for Africa Nazarene University (ANU). It describes the research design, the methods used to gather and analyze requirements, the techniques applied during system design, and the technologies selected for implementation. The chapter further explains the testing strategies that will be used to ensure system quality and reliability. It also presents the project timeline through a Gantt chart and outlines the projected budget required during system development. Ethical considerations associated with handling user data and securing the system are also discussed. The chapter concludes with a summary of how the chosen methodology supports the overall project objectives and contributes to the successful development of a functional prototype.

3.2 Research Design / Approach

The research design adopted for this project is a modified Agile Software Development approach, specifically the Agile Scrum model. This design was selected due to its iterative nature, flexibility, and suitability for student-led development projects with evolving requirements. Agile allows the system to be developed in incremental stages known as sprints, each focusing on specific features such as reporting lost items, uploading found items, automated matching, or generating reports. This approach ensures continuous refinement of the system based on user feedback.

Agile is appropriate for this project for several reasons. First, the objectives of the project involve building a user-facing mobile application that must be tested and improved gradually. Agile allows early development of partial working features that can be validated by stakeholders such as students, security officers, and administrators. Second, the scope of the project includes multiple functional components—registration, reporting, verification, notifications, and analytics—which can be developed and tested independently through sprints. Third, since the project is constrained by time and academic milestones, Agile promotes strict sprint-based deadlines, enabling the project to progress predictably.

Although traditional models such as the Waterfall approach offer structural clarity, they lack the flexibility needed to incorporate changes arising from evaluation, feedback, or new requirements. Given that stakeholder expectations may evolve during prototype development, Agile provides the adaptability and responsiveness required. The incremental delivery of features also helps reduce risk by ensuring that functional components are completed and tested continuously rather than at the end of the development cycle. For these reasons, Agile Scrum is the most suitable approach for the development of the Web-Based Lost and Found Management System.

3.3 Research Methods

The research methods for this project are structured around four key components: gathering requirements, designing the system architecture, selecting programming tools, and testing the system. Each component contributes to ensuring that the developed system aligns with the needs of users and meets the functional specifications defined in the project objectives.

3.3.1 Requirements Gathering Methods

Requirements gathering is essential for ensuring that the system addresses the actual needs of the target users—students, administrators, and campus security officers. This project employed a combination of **stakeholder interviews**, **structured questionnaires**, and **analysis of existing systems** to obtain both functional and non-functional requirements.

Stakeholder Interviews:

Interviews were conducted with five key personnel at Africa Nazarene University, including two students, one security officer, one staff representative, and one administrator. The interviews collected qualitative insights into challenges associated with current lost and found procedures, including inefficiencies in manual reporting, lack of real-time updates, and delays in item verification. Interview responses helped refine system features such as verification workflows, reporting steps, and the need for automated notifications.

Structured Questionnaires:

Questionnaires were distributed electronically to a sample of 30 students across different faculties. The questionnaire included closed-ended and open-ended questions aimed at

identifying common lost items, frequency of item loss, preferred reporting platforms, and expectations for a digital system. Responses helped quantify user preferences and determine the level of demand for system features such as image uploads, category filters, and mobile access.

Analysis of Existing Systems:
Existing platforms—both local and international—were analyzed to understand their strengths, weaknesses, and best practices. Systems from Kenya Airports Authority, NTSA, and international platforms such as Chargerback and Crowdfind were reviewed. This analysis helped identify research gaps and informed the design of a more suitable, campus-specific solution.

Data Collection and Analysis Tools:
Data collected from interviews and questionnaires were analyzed using descriptive methods. Responses were summarized in tables, charts, and thematic summaries. Qualitative data from interviews were categorized into themes such as reporting challenges, information gaps, and verification procedures. Quantitative data from questionnaires were analyzed using simple statistical measures (percentages and frequencies) to determine user expectations and requirements.

Together, these methods ensured a comprehensive understanding of user needs and allowed the system to be designed with clear functional and usability goals.

3.3.2 Design Techniques

The design phase of the system involves creating visual and structural representations of the system before actual development begins. This project employed **architectural design**, **database modeling**, and **user interface design** to build a clear blueprint for system development.

System Architecture Design:
A three-tier architecture was adopted, consisting of the presentation layer (mobile user interface), the application layer (business logic), and the database layer (data storage and retrieval). This separation enhances system maintainability, scalability, and performance. The architecture ensures that changes in the interface or business rules do not disrupt database structures.

Database Design:

A relational database model was selected for this system, implemented using SQLite for mobile storage and MySQL for the central server. Database design tools such as Entity Relationship Diagrams (ERD) were used to model entities including Users, Lost Items, Found Items, Matches, Notifications, and Reports. Proper normalization techniques were applied to reduce redundancy, improve data integrity, and support fast queries.

User Interface (UI) Design:

Figma was used to design and prototype the system's user interfaces. The design process adhered to established principles such as consistency, simplicity, visibility of system status, and error prevention. The color palette aligns with Africa Nazarene University branding (black, gold, red). Prototypes were tested with potential users to ensure usability and intuitive navigation.

Design Principles Applied:

- **Consistency:** Maintaining uniform layouts, styles, and fonts across screens.
- **User-centered design:** Interfaces were developed based on student and staff feedback.
- **Simplicity:** Clear buttons, minimal text, and straightforward processes.
- **Accessibility:** Readable contrast, responsive components, and intuitive navigation.
- **Scalability:** System designed to accommodate future additions such as QR scanning or AI-based item matching.

These design techniques ensure the system is logically structured, visually coherent, and functionally aligned with user expectations.

3.3.3 Programming Languages and Development Tools

The system is developed as a high-performance, web-based platform optimized for mobile responsiveness. This architecture ensures that the application is accessible across all devices (smartphones, tablets, and desktops) via a standard web browser, eliminating the need for platform-specific installations while maintaining a "mobile-first" user experience.

The system utilizes the following technical stack:

- **HTML5 & CSS3 (Mobile-First Responsive Design):** Used to construct a fluid and adaptive user interface. By employing a "mobile-first" philosophy, the system ensures that the UI components, navigation, and layouts are optimized for handheld devices, providing a native-app feel on mobile browsers.
- **Vanilla PHP 8+ (Backend Logic):** The primary server-side language used for processing business logic, handling user authentication, and managing data flow. PHP provides a robust and secure environment for developing custom management systems without the reliance on heavy external frameworks.
- **JavaScript (ES6):** Used for client-side interactivity, form validation, and enhancing the user experience through asynchronous updates, ensuring the system remains fast and responsive during user sessions.
- **MySQL Database:** Acts as the centralized relational database management system. It securely stores all data related to user accounts, item reports, and claim logs, ensuring high data integrity and efficient retrieval.
- **Bootstrap 5 (UI Library):** A professional front-end toolkit used to ensure a clean, modern aesthetic with standardized components (cards, modals, and buttons) that are fully compatible with the Africa Nazarene University (ANU) branding guidelines.
- **Progressive Web App (PWA) Technology:** The system is structured to support PWA features, allowing students and staff to "Add to Home Screen" for instant access. This provides a shortcut and an app-like experience for reporting or searching for items quickly while on the move.
- **Visual Studio Code (IDE):** The primary development environment used for writing, debugging, and managing the source code.
- **XAMPP Server:** Used as the local development stack to host the Apache web server and MySQL database during the implementation and testing phases of the project.

This selection of technologies ensures a lightweight, scalable, and highly accessible solution that meets the specific administrative and security needs of the University.

3.3.4 Testing Methods

System testing is crucial to ensure that the application performs correctly and meets user expectations. The following testing methods will be used:

Unit Testing:
Each module—such as user registration, reporting lost items, or verification—will be tested individually. The goal is to verify that each component works as expected in isolation.

Integration Testing:
After individual components are successfully tested, they will be integrated to evaluate data flow and interaction. For example, integration tests will confirm that lost item reports submitted from the mobile app are stored in MySQL via the backend API.

System Testing:
This method tests the entire application as a unified system. It includes UI testing, backend validation, and database testing. The objective is to verify that all components work together seamlessly.

User Acceptance Testing (UAT):
UAT will be conducted with a small group of students, administrators, and security officers. They will use the system and provide feedback on usability, performance, and accuracy. Issues identified will be fixed before final deployment.

Performance Testing:
The system will be tested under simulated load conditions to ensure it can handle multiple user requests without lag or crashes.

Security Testing:
This includes checking for login vulnerabilities, data protection, secure API communication, and role-based access control.

Testing ensures system reliability, robustness, and alignment with project goals.

3.4 Timeline (Gantt Chart)

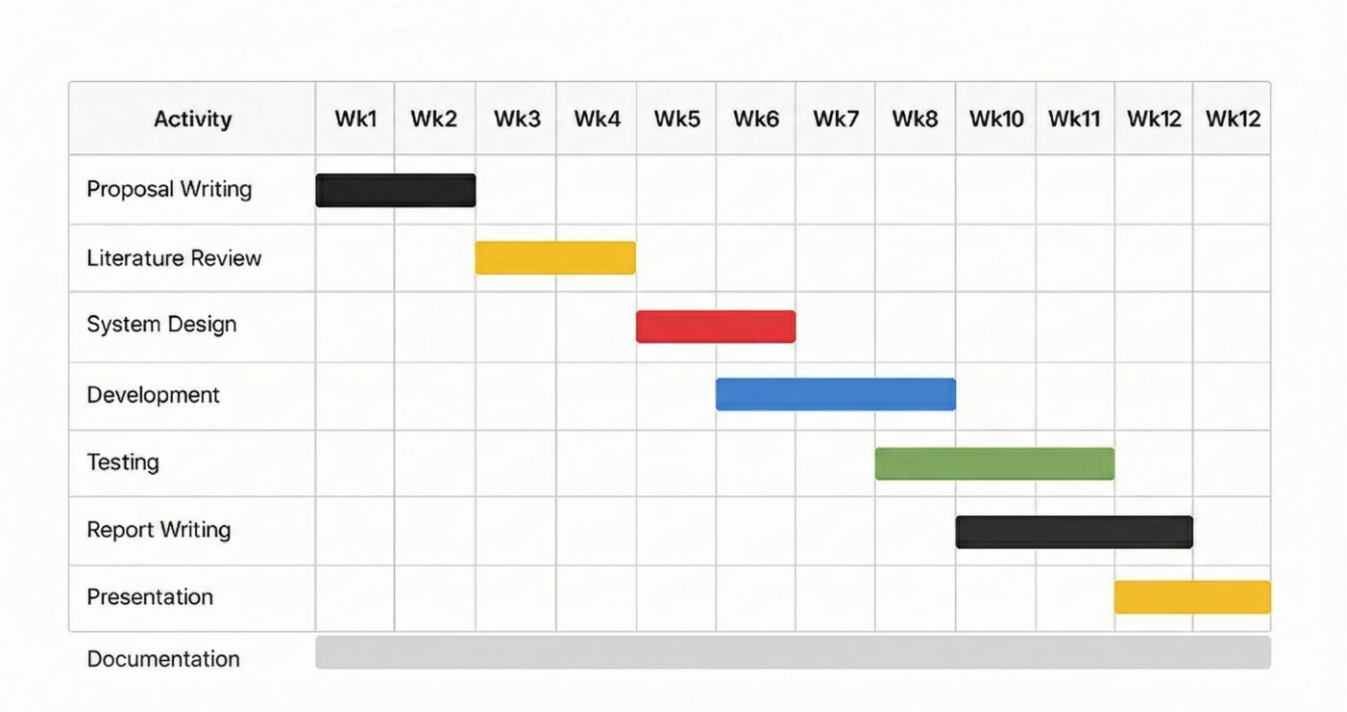

Figure 1: Gantt Chart

3.5 Budget

Item	Description	Cost (KES)
Internet bundles	Research, development, deployment	50,000
Software tools	Figma, hosting, GitHub utilities	35,000
Transport	Stakeholder interviews	10,000
Miscellaneous	Printing, materials	5,000
Total Estimated Cost		**100,000**

Table 1: Budget Breakdown

3.6 Ethical Considerations

Ethical considerations for this project focus on data privacy, secure handling of user information, compliance with relevant university and ICT policies, and responsible system behavior.

- **Confidentiality:** Personal information such as student IDs, names, and contact details will be stored securely and accessed only by authorized personnel.
- **Privacy:** Users will be informed about how their data will be used, and consent will be obtained before account creation.
- **Data Security:** Encryption, secure passwords, and role-based access control will be implemented.
- **Accuracy:** Matching algorithms will be transparent and documented to avoid false positives.
- **Fair Use:** The system must not be used to profile, track, or unfairly target users.
- **Compliance:** The system adheres to ANU ICT policies, Kenyan Data Protection Act, and ethical research guidelines.

3.7 Conclusion

This chapter has outlined the methodology used to guide the development of the Web-Based Lost and Found Management System for Africa Nazarene University. The adoption of an Agile approach ensures iterative development, flexibility, and incorporation of user feedback throughout the process. Detailed requirements gathering methods ensured that the system is grounded in actual user needs, while design techniques and programming choices ensure scalability, usability, and maintainability. Testing strategies guarantee reliability and quality of the final system. The inclusion of a clear timeline, budget, and ethical considerations strengthens the feasibility and integrity of the project. Overall, the methodology aligns closely with the project's objectives and supports the creation of an effective, user-centered, and functional system prototype.

CHAPTER FOUR: SYSTEM DESIGN

4.1 Functional Requirements

Functional requirements define the core features and operations that the Web-Based Lost and Found Management System must perform to meet the objectives of the project. These requirements were derived from stakeholder interviews, questionnaires, and analysis of existing systems. They represent the essential functional behavior of the system from the perspective of each user group: students, security officers, and administrators.

Summary of Functional Requirements

1. **User Registration & Login**
    - The system must allow students, administrators, and security officers to register using their ANU credentials.
    - Users must authenticate using a secure login form.
2. **Report Lost Item**
    - Students must be able to submit details of a lost item, including category, description, date lost, last known location, and an optional photo.
3. **Report Found Item**
    - Any user (students/security officers) must be able to record found items with item description, category, location found, and photo.
4. **Automated Matching of Lost & Found Items**
    - The system must automatically compare lost and found items through keyword matching, category similarity, and date/location proximity.
    - When a match is detected, the system should generate a notification.
5. **Notifications & Alerts**
    - Users should receive alerts when:
        - Someone reports an item similar to what they lost

        - A security officer verifies an item
        - Their lost item has been recovered

6. **Verification & Approval**
    - Security officers must be able to confirm or reject potential matches.
    - Verification includes viewing the reporter's details and item image.
7. **Item Recovery & Status Update**
    - Once an item is collected by its owner, the status must be updated to "Recovered".
8. **Report Generation**
    - Administrators must generate monthly or weekly summaries of lost, found, and recovered items.

4.1.1 Use Case Diagram

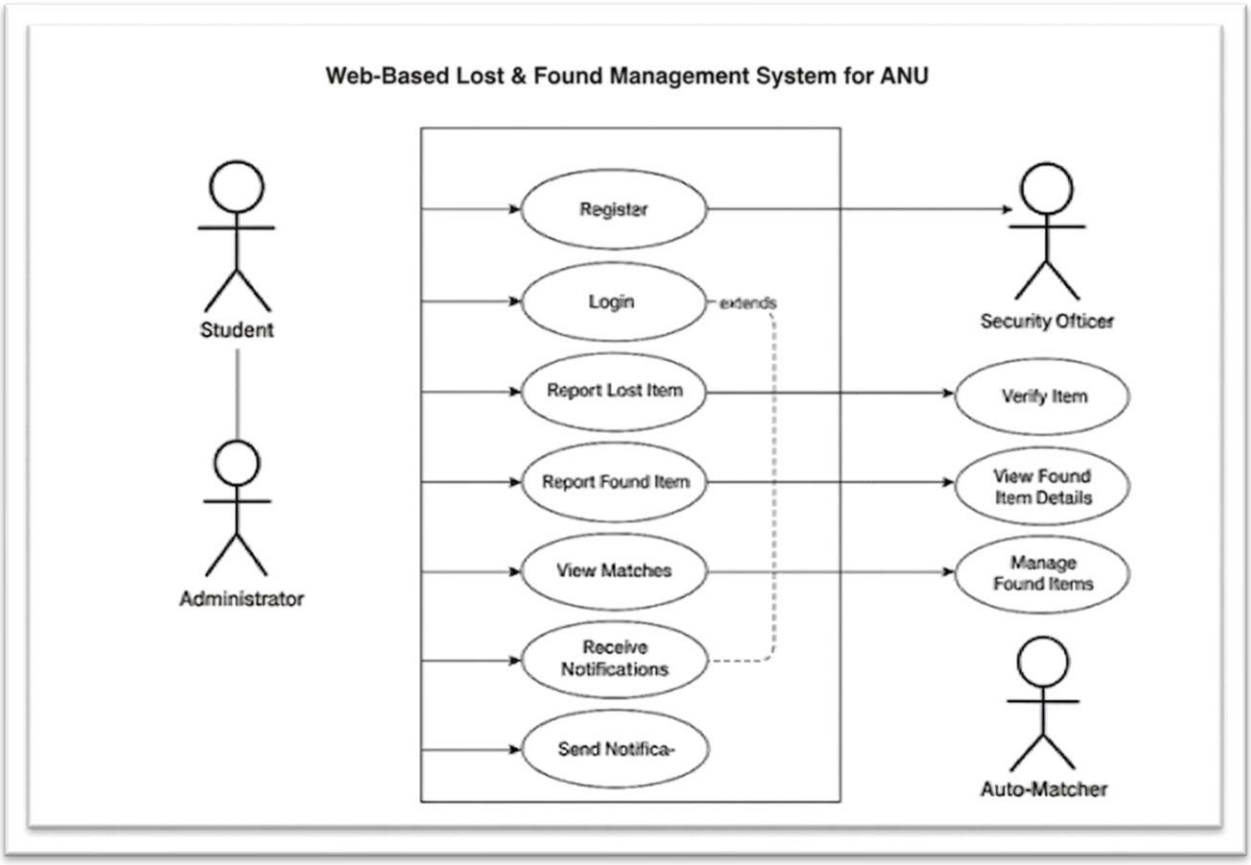

Figure 2: Use Case Diagram

4.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance constraints, and operational characteristics of the system.

1. Performance Requirements

- The system must load dashboard screens within **3 seconds**.
- Matching operations must execute in **under 2 seconds** for up to 500 records.

2. Scalability Requirements

- The system must support growth to **5,000 users** without performance degradation.
- Database structure must allow new item categories and new user roles in the future.

3. Security Requirements

- Passwords must be encrypted using SHA-256 hashing or equivalent.
- All communication between app and API must use HTTPS.
- Access control must follow role-based authorization.

4. Reliability Requirements

- System uptime target: **99%**.
- Automatic backup of MySQL database every 24 hours.

5. Usability Requirements

- Interface must follow simplicity principles and use ANU-themed color palette.
- A new user should complete item reporting in **less than 2 minutes**.
- The system must support visually clear buttons and readable fonts.

6. Maintainability Requirements

- Code must be modular with distinct components (auth, reporting, notifications).
- Developer documentation must accompany API endpoints.

7. Portability Requirements

- The web application must be compatible with all modern web browsers.
- Backend must be deployable on any Linux-based server.

4.3 Context Diagram

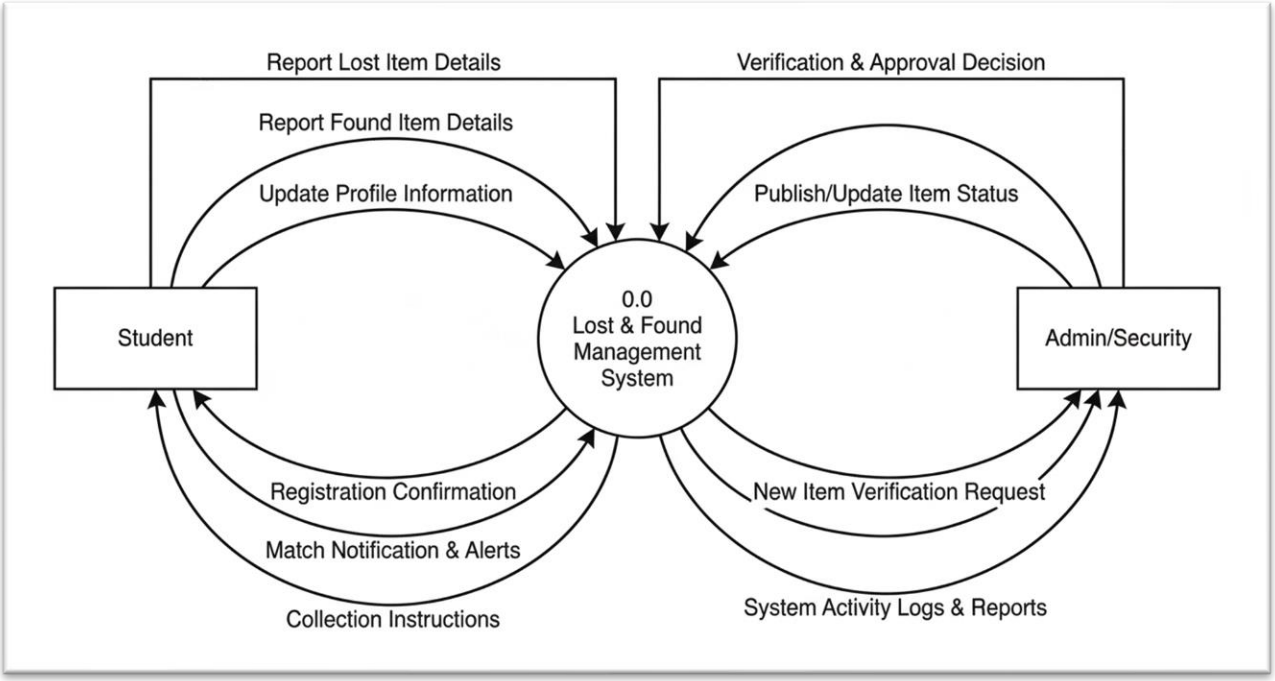

Figure 3: Context Diagram

4.4 Data Flow Diagram (DFD) – Level 0

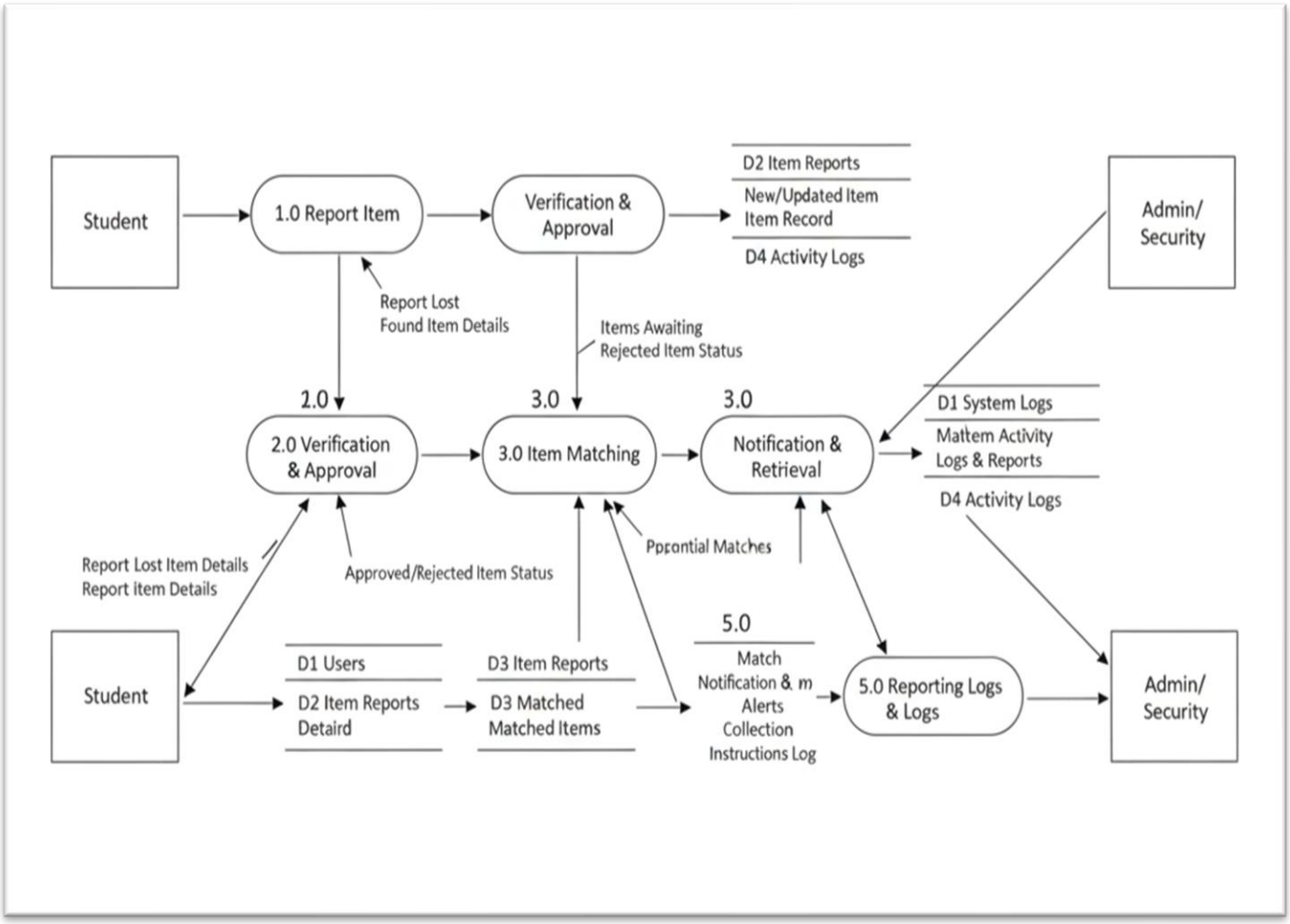

Figure 4: Data Flow Diagram - Level 0

4.5 Class Diagram

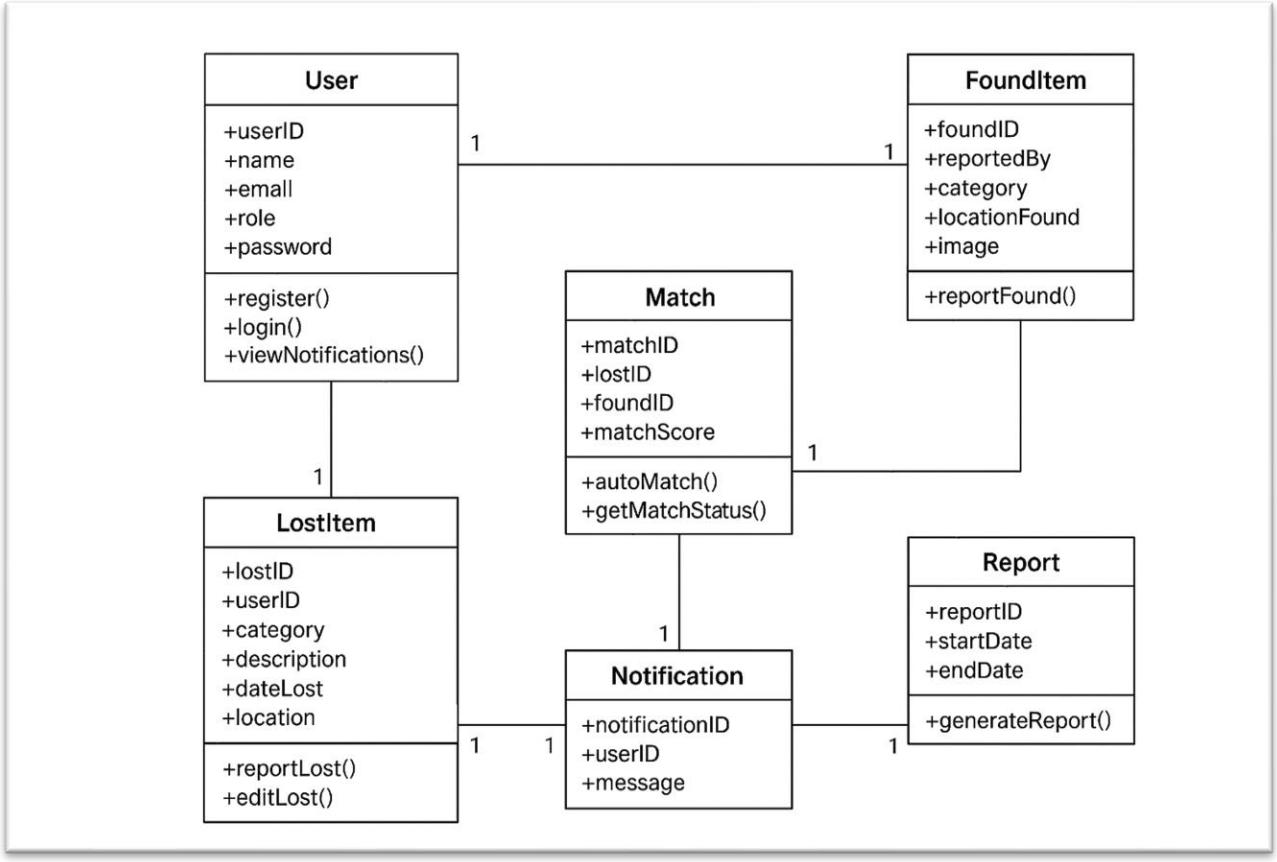

Figure 5: Class Diagram

4.6 Entity Relationship Diagram (ERD)

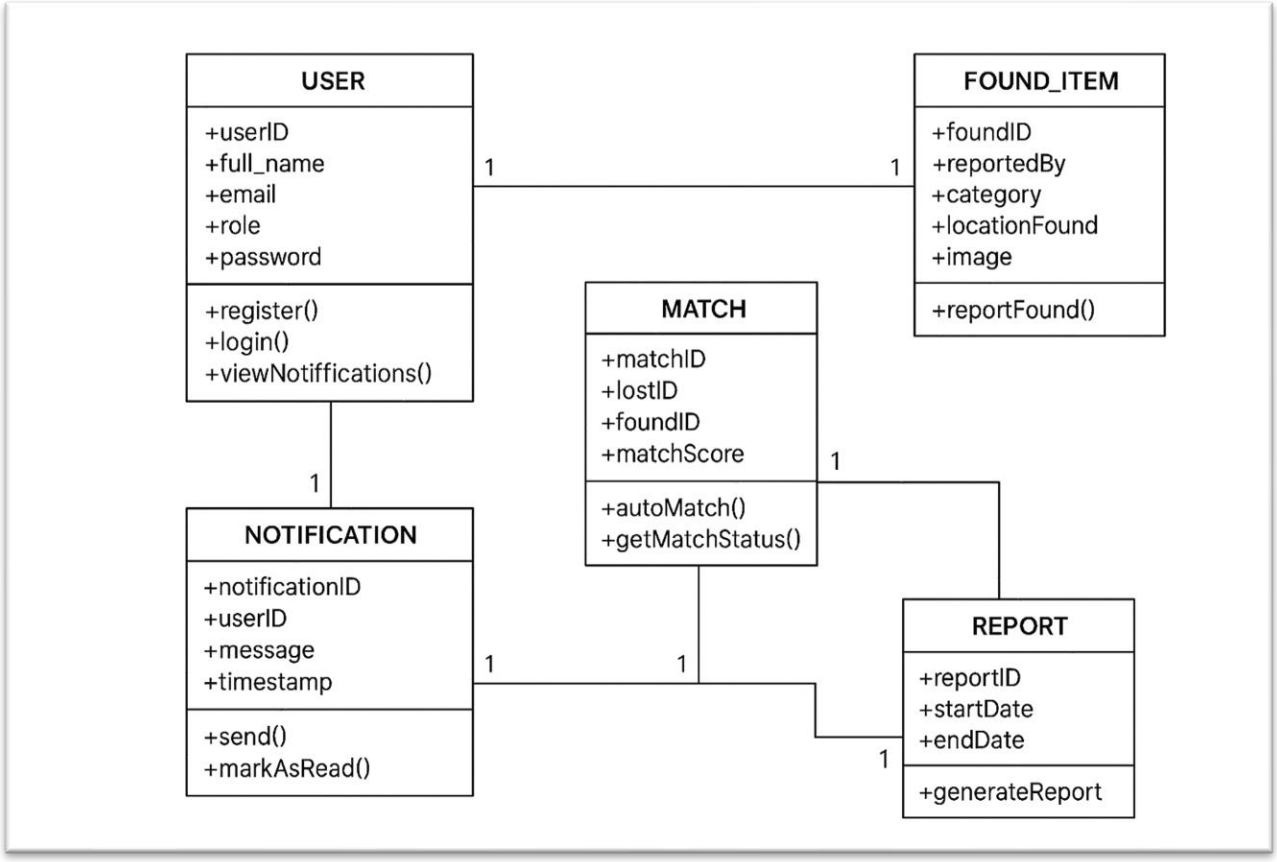

Figure 6: Entity Relationship Diagram

CHAPTER FIVE: SYSTEM PROTOTYPE, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the working prototype of the Web-Based Lost and Found Management System developed for Africa Nazarene University (ANU). It explains how different users interact with the system and demonstrates how the system meets the objectives outlined in Chapter One. Screenshots of the prototype are provided to illustrate key interfaces, workflows, and system capabilities. The chapter concludes with a summary of the project achievements and recommendations for future system improvement.

5.2 System Prototype

The Web-Based Lost and Found Management System was successfully developed as a functional prototype using HTML, CSS, JavaScript for the web interface, PHP for backend logic, and MySQL for centralized storage. The prototype implements all key features described in Chapter Four, including item reporting, automated matching, verification workflows, notifications, and administrative reporting.

The system includes three user roles—Students, Security Officers, and Administrators—each with customized dashboards and functions. The prototype demonstrates how users can report lost and found items, upload images, receive match alerts, and manage verification processes through a clean, ANU-themed interface.

Screenshots of the working prototype are provided below to illustrate core system components.

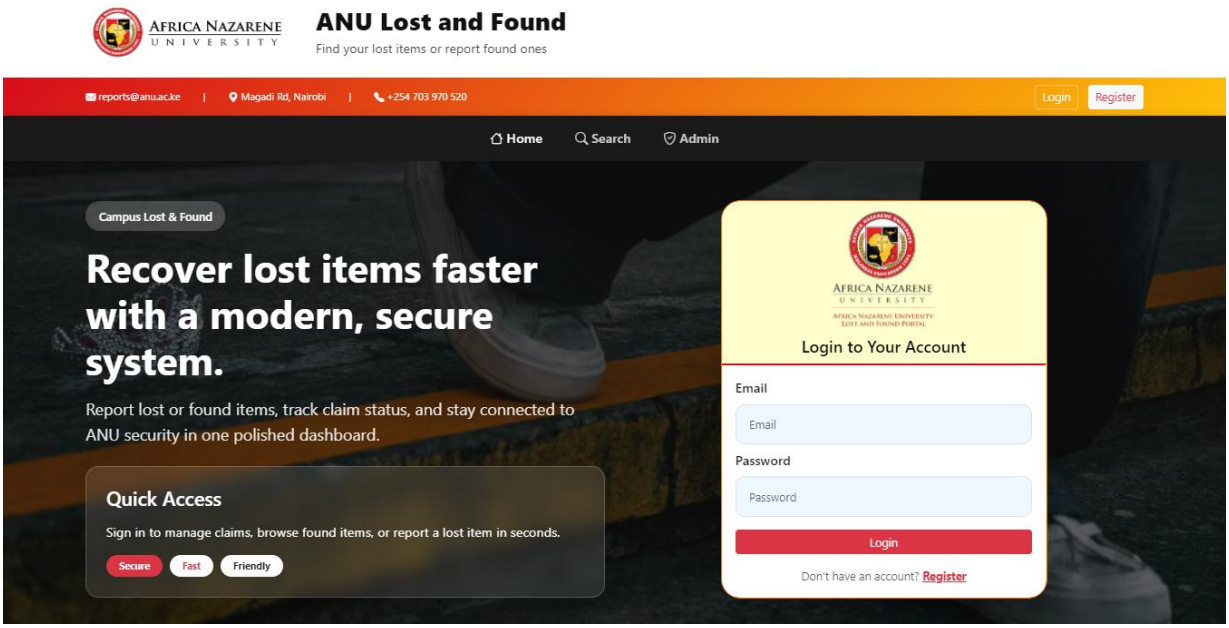

Figure 7: Login Interface

Caption:

The login screen allows students, security officers, and administrators to authenticate using their ANU credentials. The interface includes secure input fields with validation checks.

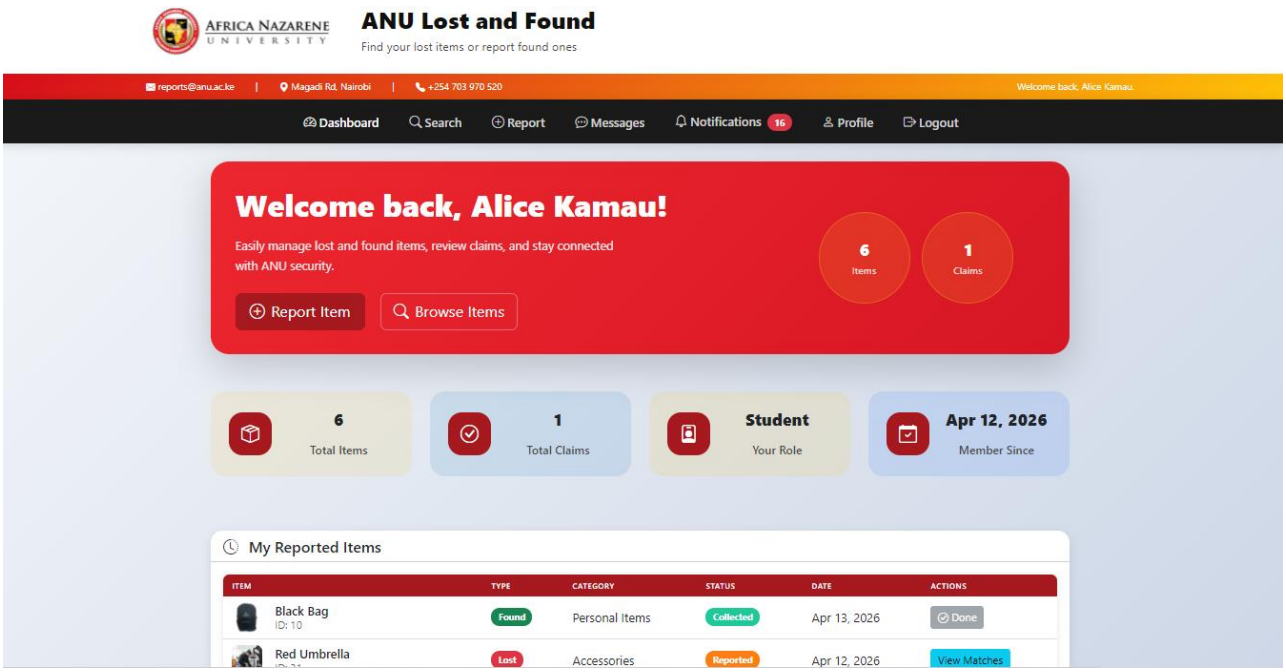

Figure 8: Student Dashboard

Caption:

The student dashboard provides quick access to reporting lost items, reporting found items, viewing matches, checking notifications, and managing profile details.

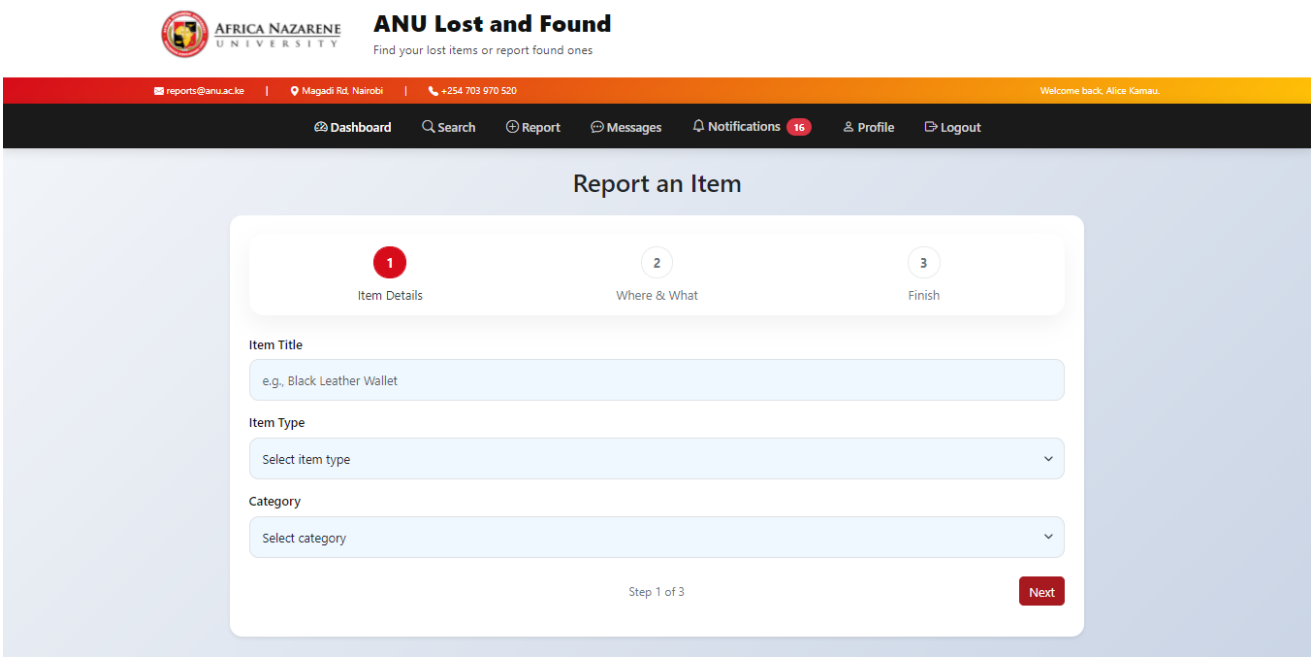

Figure 9: Lost Item Reporting Form

Caption:

Students can report lost items by entering item details, selecting the category, choosing date and location, and uploading a supporting photo.

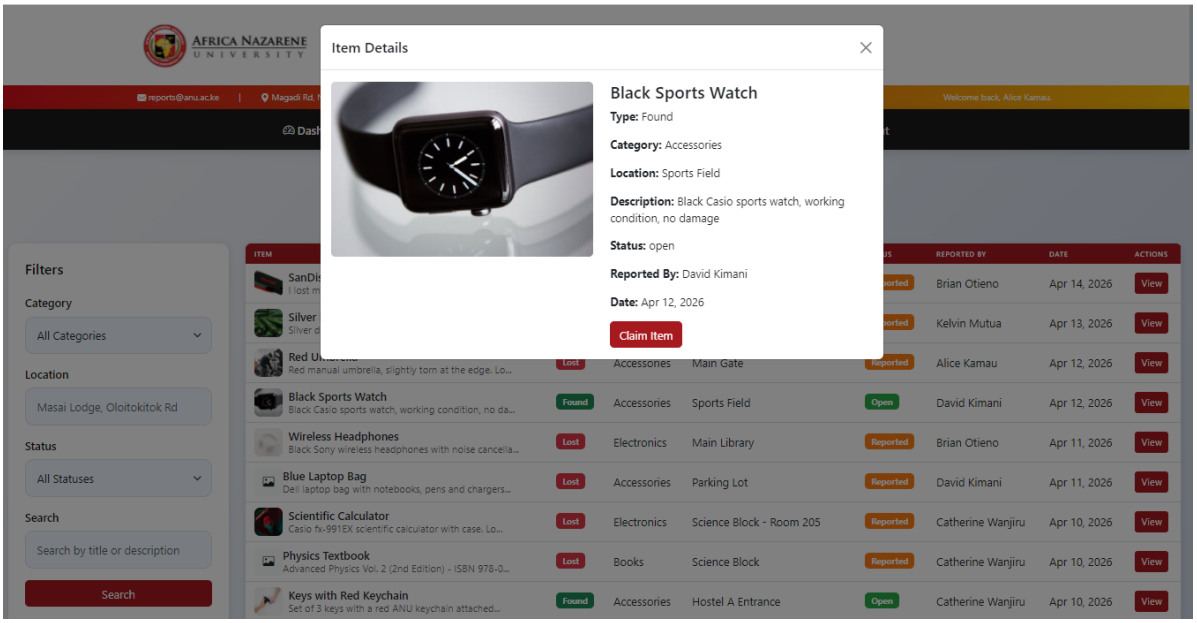

Figure 10: Match Results Interface

Caption:

The system displays automatically matched items with similarity scores. Students can confirm whether the matched item belongs to them.

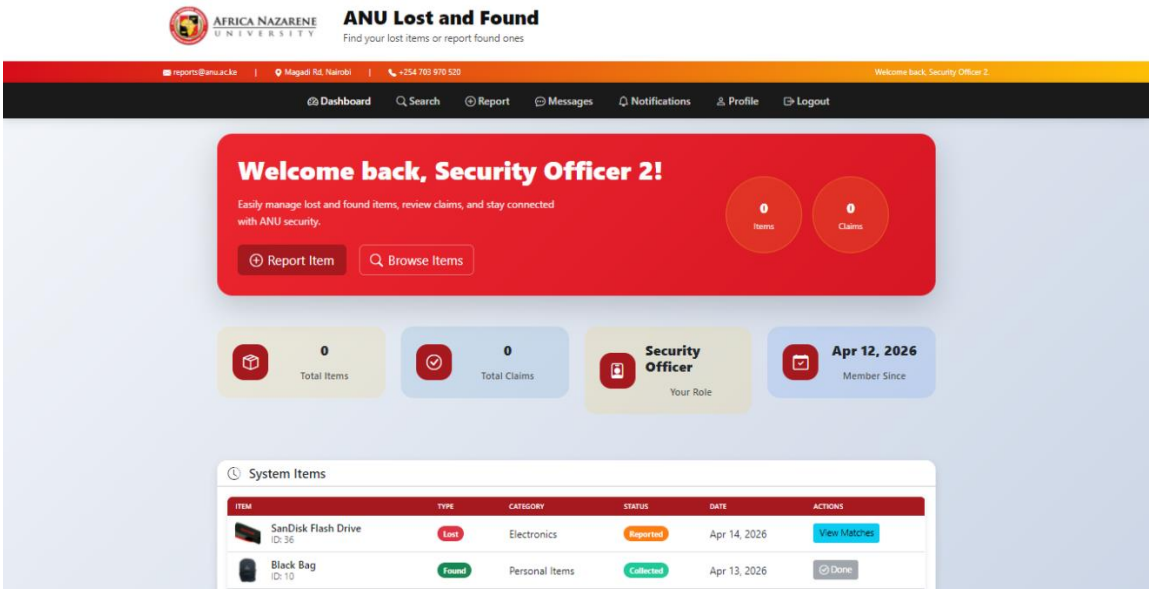

Figure 11: Security Officer Dashboard

Caption:

Security officers receive verification requests for matched items and can approve or reject ownership claims based on evidence.

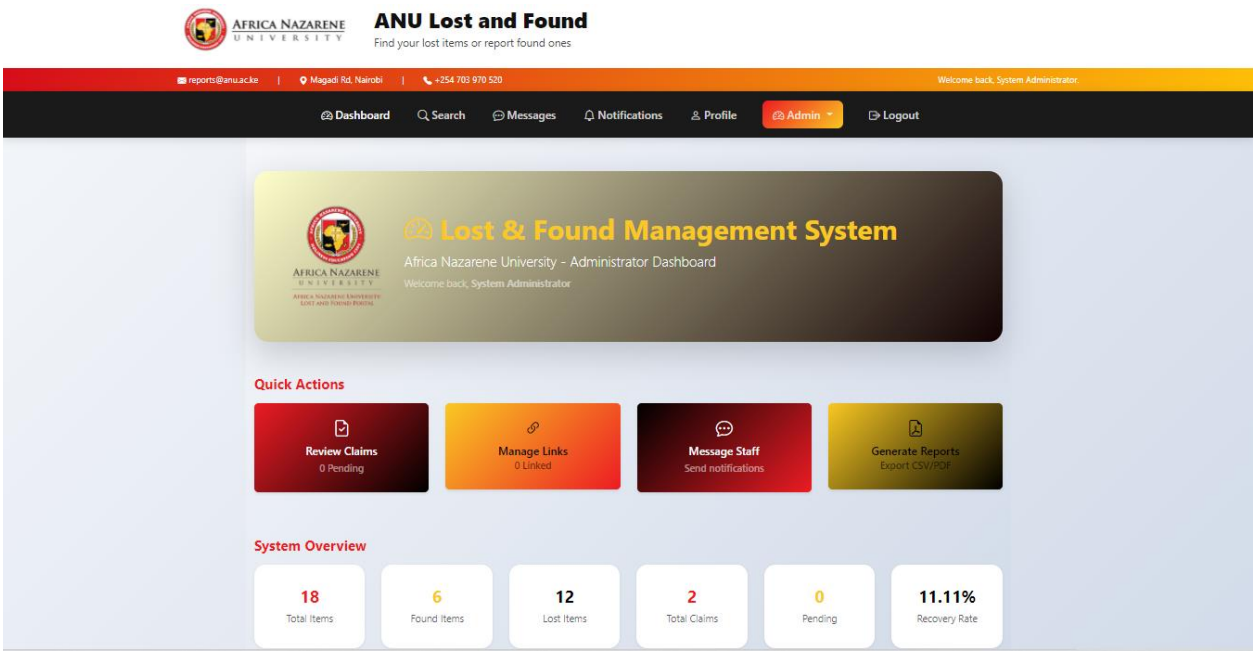

Figure 12: Admin Dashboard

Caption:

Administrators oversee all system data and can generate weekly or monthly summary reports based on lost, found, and recovered items.

5.3 System Description

The Web-Based Lost and Found Management System is a structured digital platform that digitizes item reporting and recovery processes. The mobile application enables students, security officers, and administrators to access role-specific features.

Technologies Used

- **HTML, CSS, JavaScript:** Mobile app interface and logic
- **PHP (to be used at full implementation):** Backend processing and communication
- **MySQL Database:** Central server-side storage for lost items, found items, users, matches, and reports
- **SQLite:** Local storage for offline caching
- **Figma:** UI/UX prototyping and interface blueprint
- **JSON:** Data formatting for communication between APIs

Data Flow

The system retrieves and stores records through a central MySQL database. When implemented fully, it will synchronize offline data stored in SQLite with the server once internet connectivity is available.

This description aligns with the architecture and design principles presented in Chapter Four.

5.4 User Interaction

This section details how each user role interacts with the system. Screenshots are provided to show step-by-step user actions and how the system responds.

5.4.1 Student Interaction

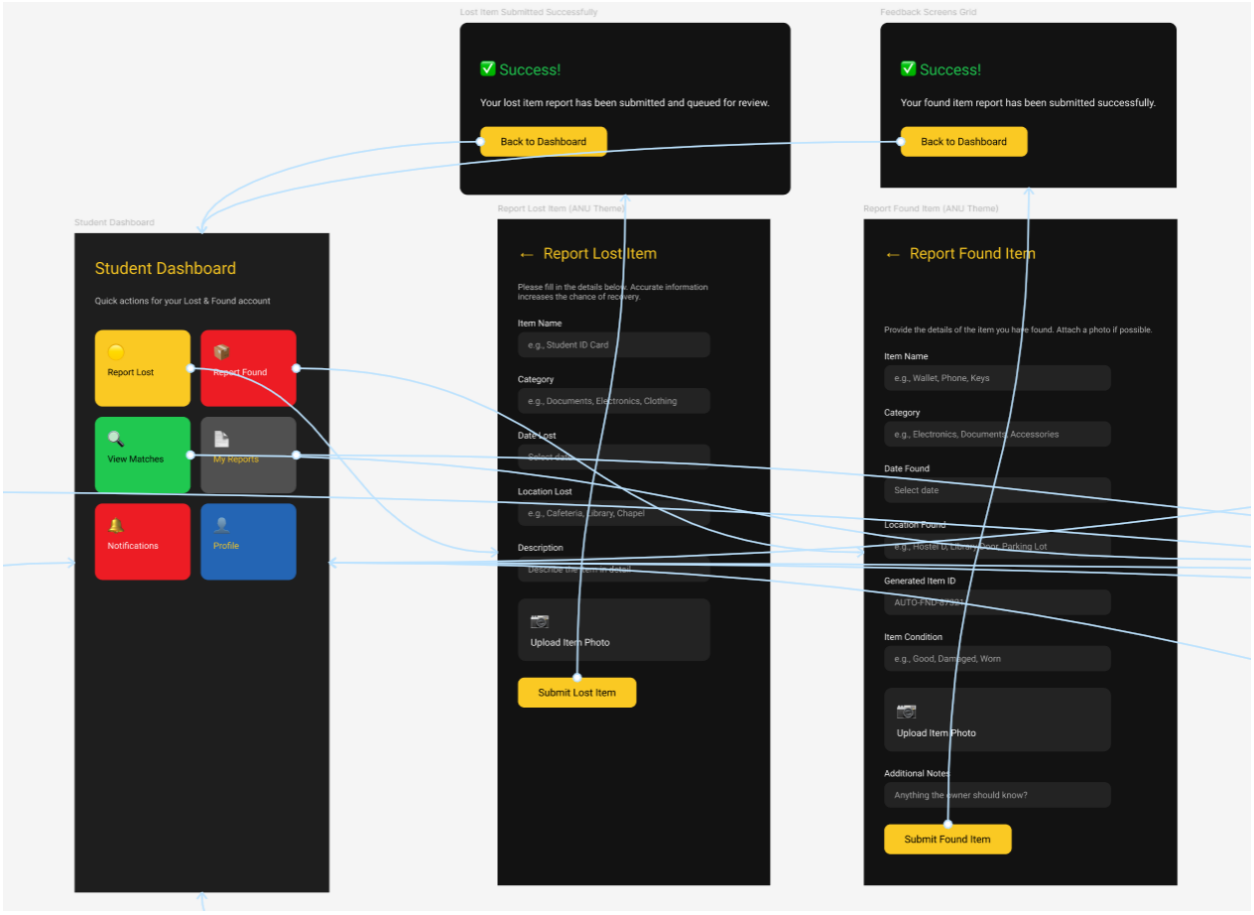

Figure 13: Student Reporting a Lost Item

Explanation:

This screen allows students to enter item details and upload photo evidence. Once submitted, the system stores the data and begins automatic matching.

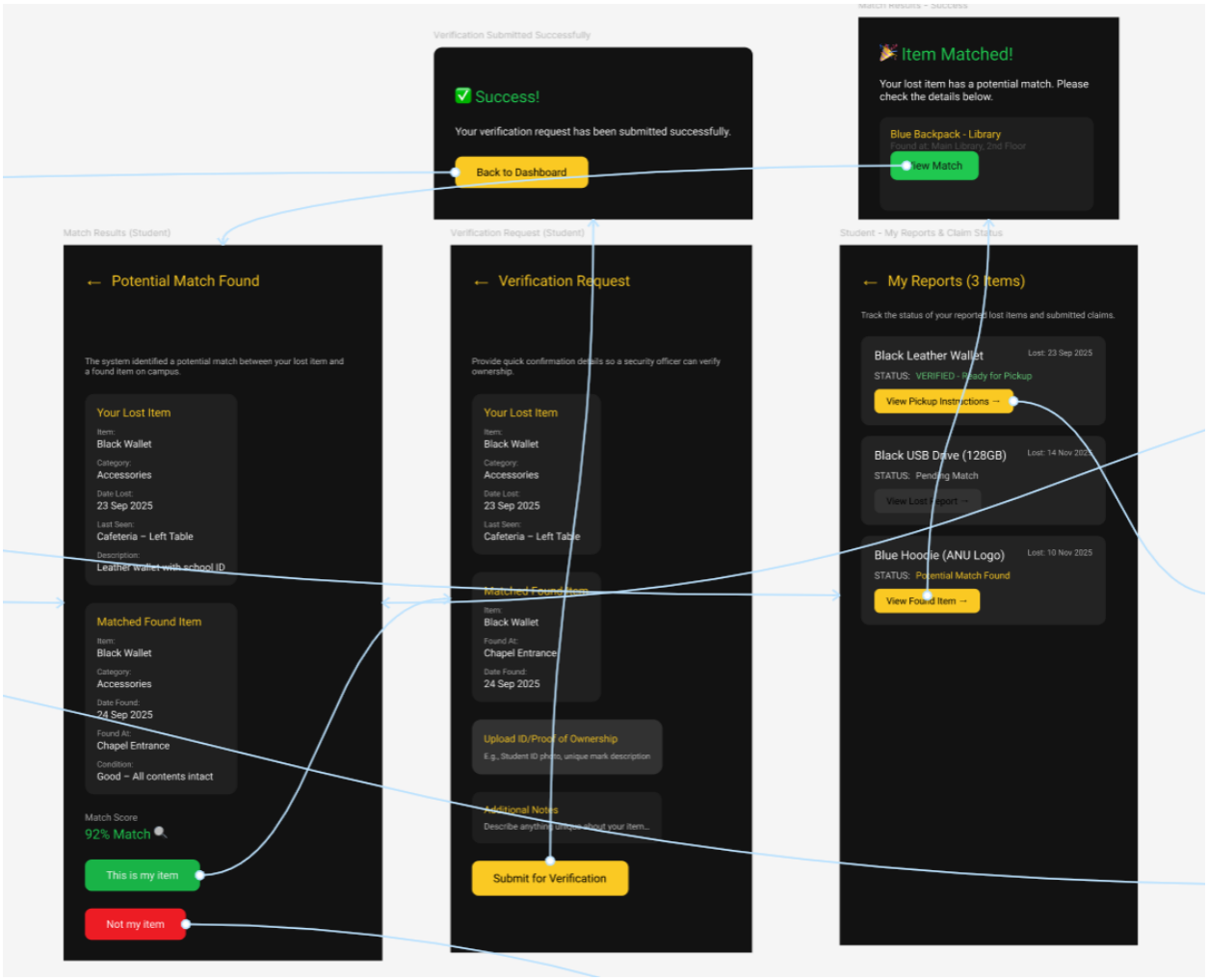

Figure 14: Student Viewing Match Notifications

Explanation:

When a potential match is found, the student receives a real-time in-app notification. They can view matched items and confirm ownership.

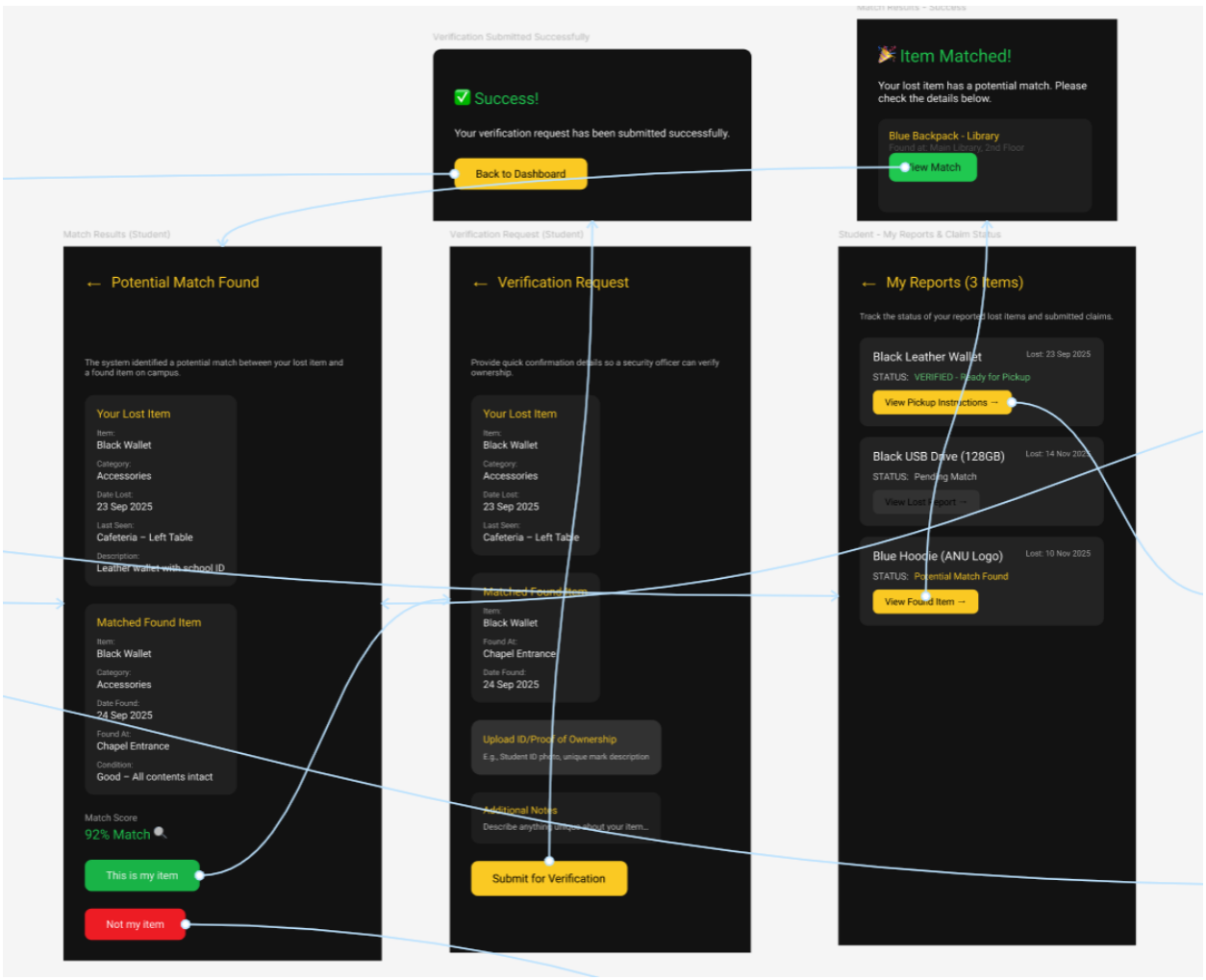

Figure 15: Student Confirming a Match

Explanation:

Students verify whether the suggested found item belongs to them. If confirmed, the case is forwarded to a security officer for final approval.

5.4.2 Security Officer Interaction

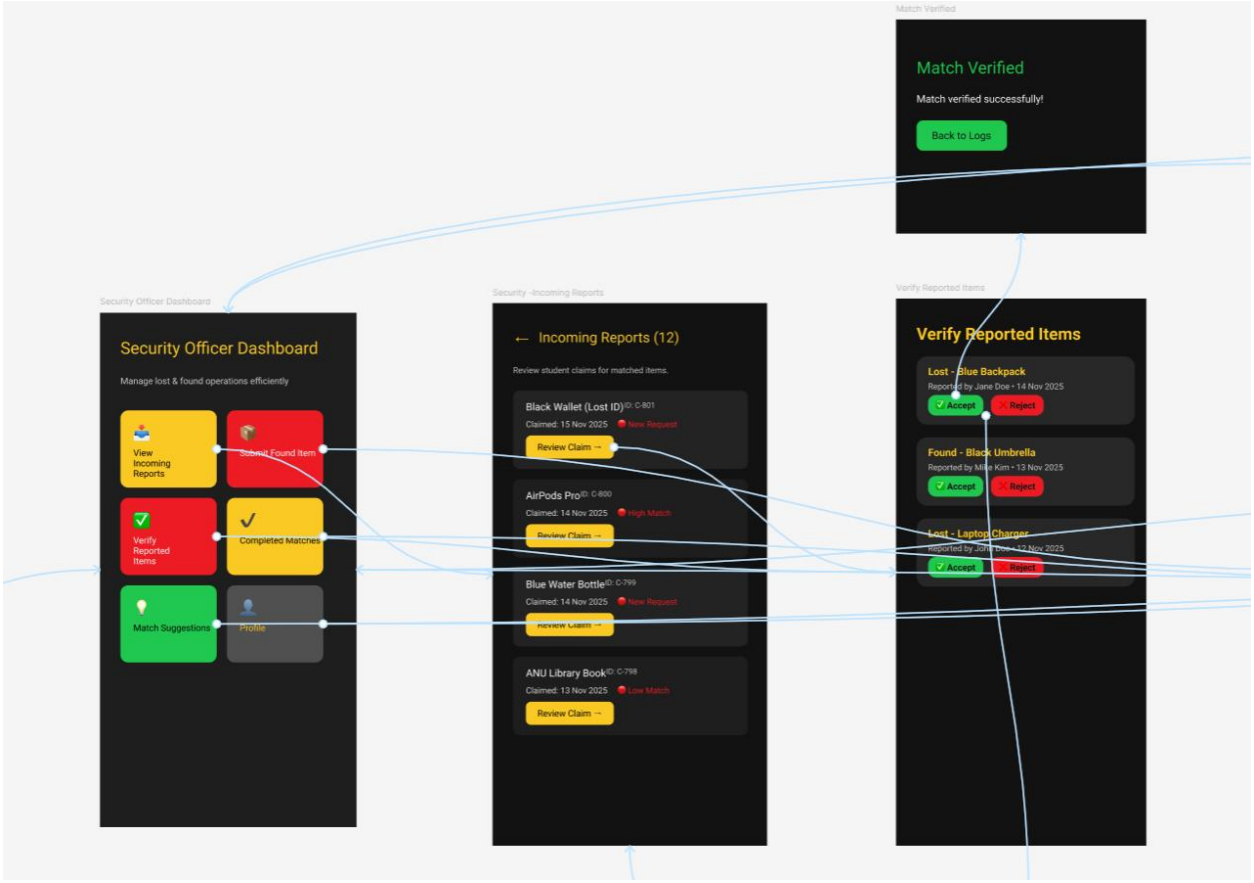

Figure 16: Verification Queue

Explanation:

Security officers see all pending verifications, each containing the reported lost item and the matched found item.

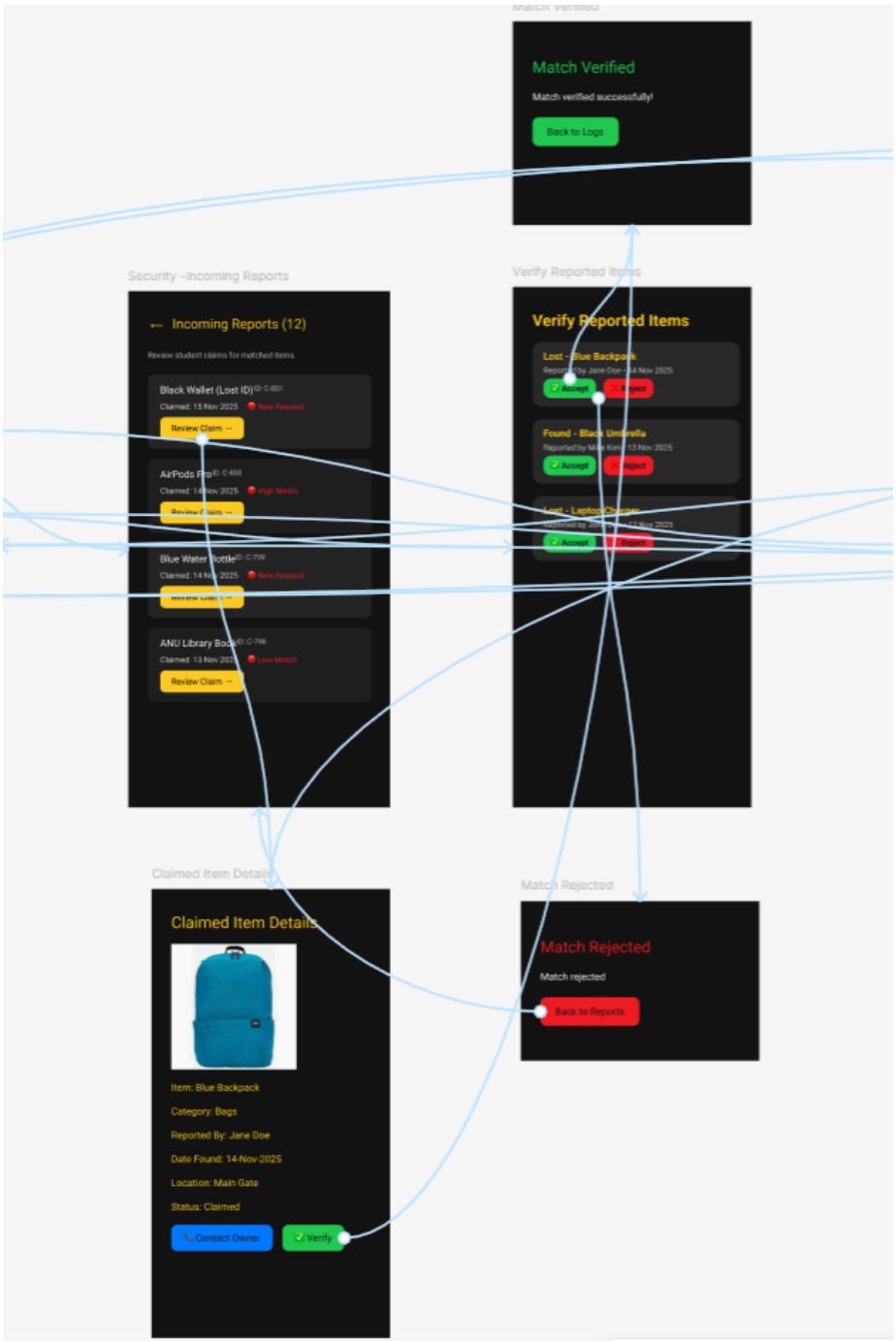

Figure 17: Verification Decision Screen

Explanation:

The officer reviews images, details, and reporter information, then approves or rejects the match.

5.4.3 Administrator Interaction

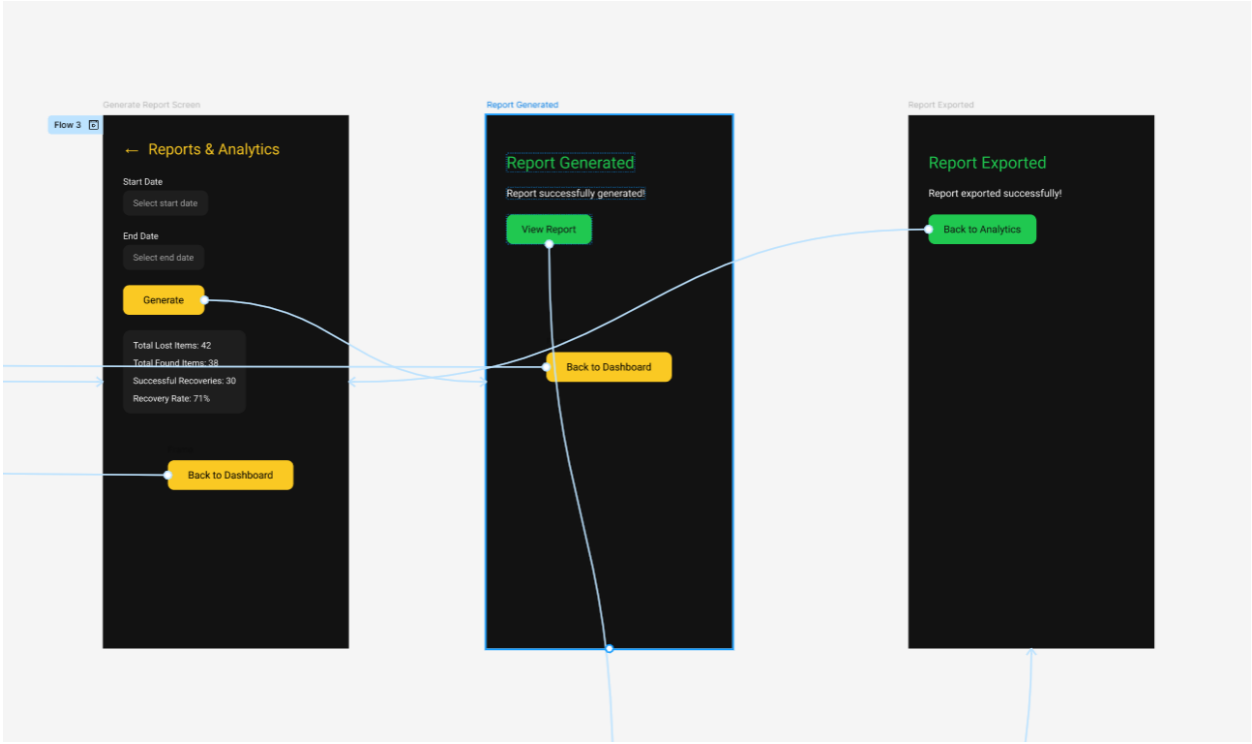

Figure 18: Reports Generation Interface

Explanation:

Administrators select a date range to generate reports summarizing lost, found, matched, and recovered item statistics.

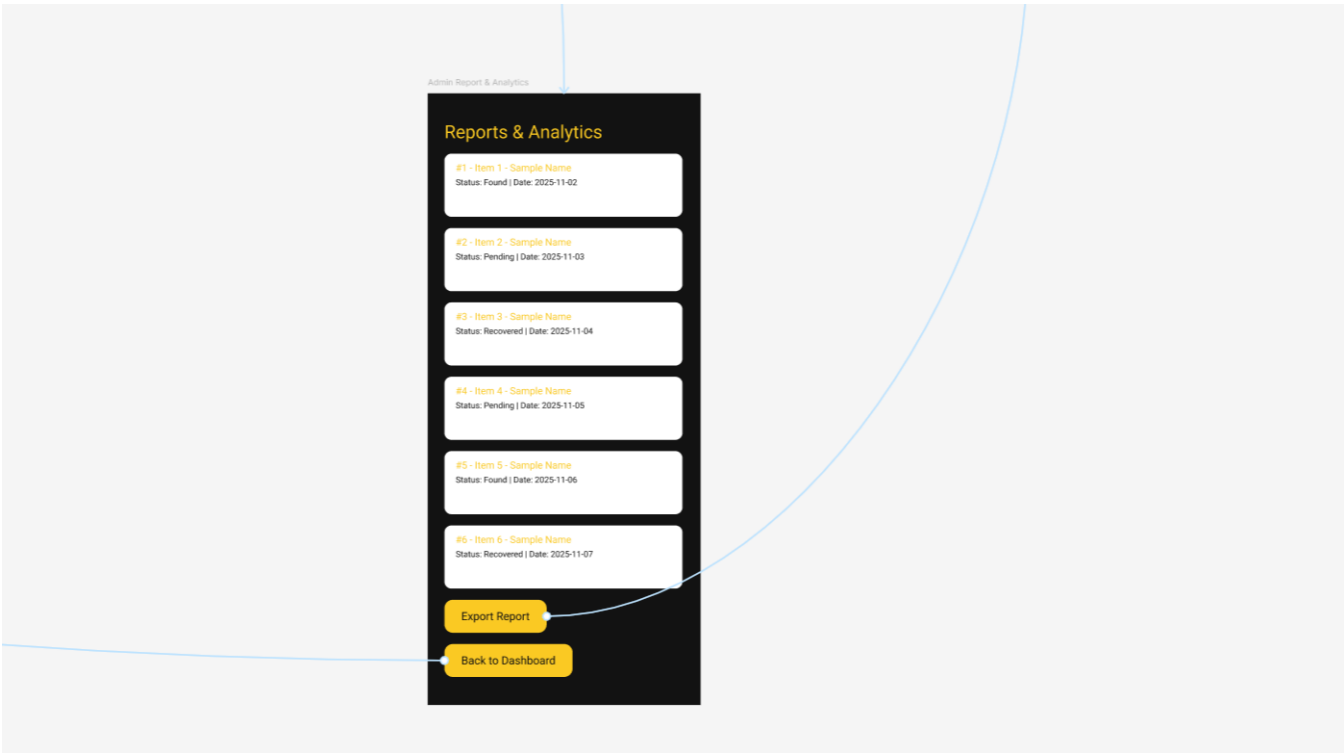

Figure 19: Generated Summary Report

Explanation:

The generated report includes insights on recovery trends, match rates, and activity summaries, helping administrators evaluate system effectiveness.

5.5 Demonstration of Core Functionalities (Against Project Objectives)

Below is a structured comparison showing how each objective listed in Chapter One is achieved by the system prototype:

Objective (from Chapter One)	How the Prototype Achieves It
1. To enable students to create and manage personal accounts for reporting and tracking lost or found items within the university.	The prototype includes the Login Screen, Role Selection, and Account Registration prompts. It demonstrates secure user entry, account access, and a dashboard where users can track reported items.
2. To develop a module that allows users to report lost or found items by capturing details such as item category, description, photos, and location.	The prototype contains structured "Report Lost Item" and "Report Found Item" screens with form fields for item category, description, date, photo upload placeholders, and location inputs.
3. To implement an admin/security verification system for reviewing, approving, and publishing reported items to ensure accuracy and prevent false reports.	Admin/Security screens in the prototype include item verification pages, approval buttons, and lists of pending reports — demonstrating how administrators review and publish valid items.
4. To design an automated matching engine that compares lost and found item reports using keywords and categories, and alerts users when a potential match is identified.	The prototype includes a dedicated "Match Items" screen with sample matched items. Status indicators (e.g., "Matched") and dashboard alerts simulate how the matching engine notifies users.
5. To provide real-time notifications to students regarding item matches, verification updates, and collection instructions.	The dashboard features a Notifications section and item status updates showing real-time alerts such as "Verification Pending," "Item Matched," and "Ready for Collection."
6. To generate administrative reports showing matched items, recovery rates, frequently lost items, and system activity logs to support decision-making.	A "Generate Reports" screen displays report categories, sample summaries, and placeholders for charts and logs — illustrating how administrative reporting is handled.

Table 2: Comparison of core functionalities against project objectives

Prototype Link:

https://www.figma.com/proto/Z33Tx5x11qJXzCdUN42foj/Mobile-Lost---Found-Management-System-for-Africa-Nazarene-University--ANU-

5.6 Conclusion

This chapter presented the functional prototype of the Web-Based Lost and Found Management System for ANU. The prototype successfully demonstrates how digital reporting, automated matching, security verification, and administrative reporting can improve item recovery efficiency within the university. The system addresses the initial problem of slow and fragmented manual processes by offering structured reporting, real-time notifications, and digital verification. Based on the user flows and implemented interfaces, the prototype meets all the objectives outlined in Chapter One. Throughout development, challenges such as interface consistency, workflow alignment, and data structuring were addressed through iterative design and feedback. The system provides a strong foundation for future deployment and enhancements.

5.7 Recommendations

The following recommendations can guide future improvements and full system implementation:

1. **Integration with AI-based image recognition** to improve the accuracy of automated matching.
2. **QR code tagging** for frequently lost items such as student IDs and library cards.
3. **Push notifications**
4. **Offline-first mode** allowing users to record reports without internet connectivity.
5. **Integration with university security systems** and campus access logs.
6. **Full integration with ANU's student portal** for seamless authentication.

These enhancements will strengthen system efficiency and expand usability across the university.

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Appendix A: System Prototype

This appendix presents the visual components of the Web-Based Lost and Found Management System prototype developed for Africa Nazarene University (ANU). The screenshots demonstrate the system's interface, navigation structure, and the core functionalities described in Chapters Four and Five. Each screen reflects the system's objectives, user requirements, and ANU-themed design standards, as implemented in the Figma prototype.

The prototype consists of interfaces for all three system roles—Students, Security Officers, and Administrators—and showcases how each user interacts with the system to report items, verify matches, receive notifications, and generate administrative reports.

A.1 User Interface Screens

Below is the list of UI screens to include in this appendix.

---

1. Login Screen

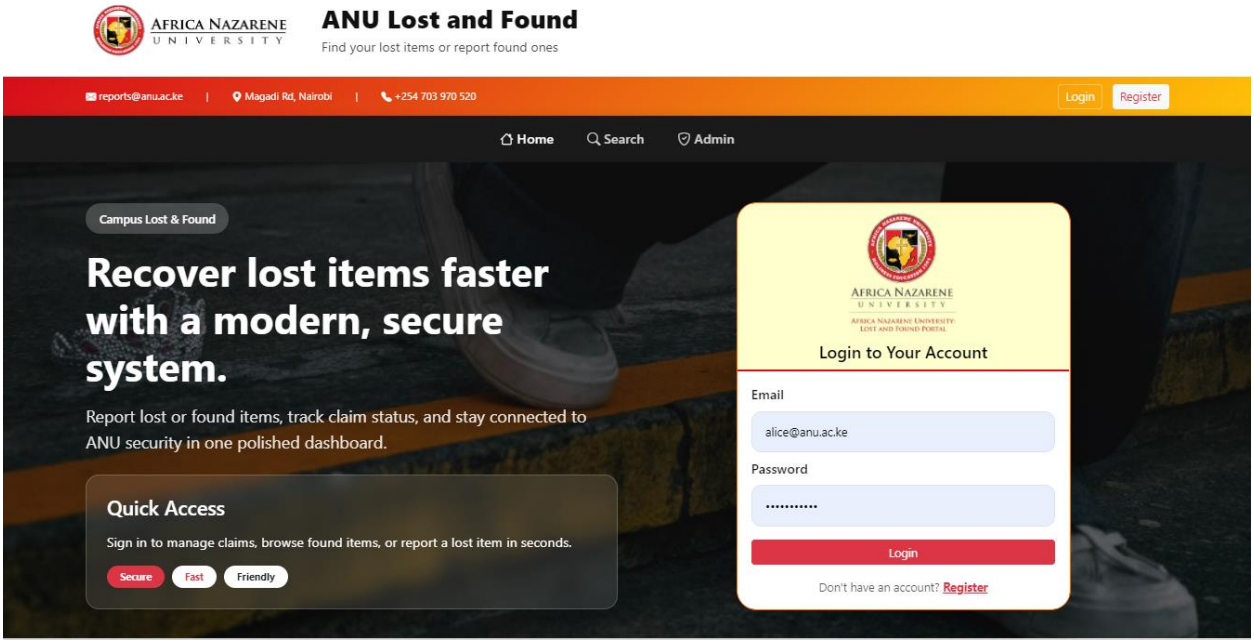

Figure 20: User Login Interface

Description:

This screen allows students, security officers, and administrators to authenticate using their ANU credentials. Users select their role and proceed to access their respective dashboards.

---

2. Student Dashboard

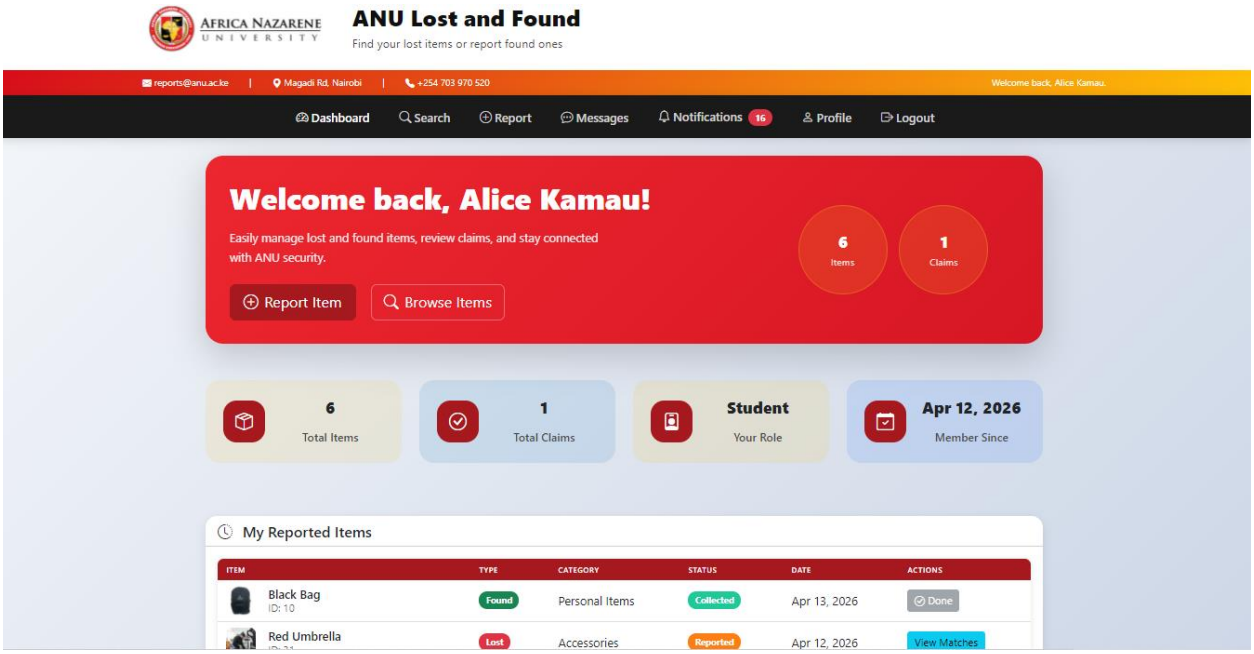

Figure 21: Student Dashboard Interface

Description:

Shows the main functions available to students, including reporting lost or found items, viewing matches, checking notifications, and accessing their profile.

---

3. Report Lost Item Screen

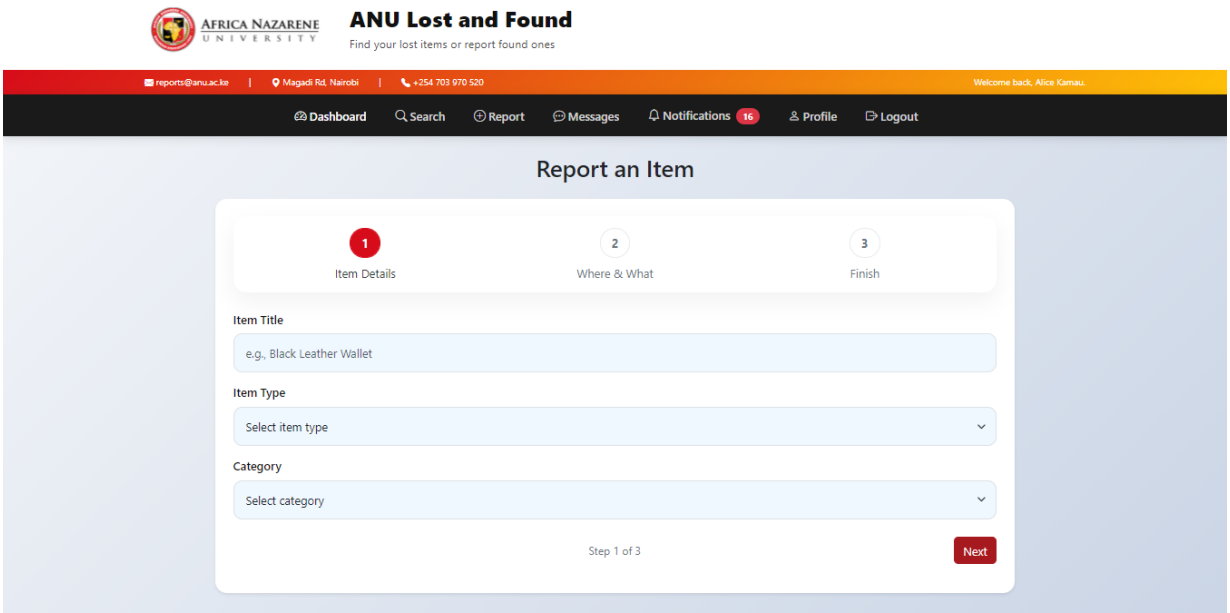

Figure 22: Lost Item Reporting Form

Description:

Students enter details about the lost item, including name, category, location, date, description, and photo upload.

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5. Match Results Screen

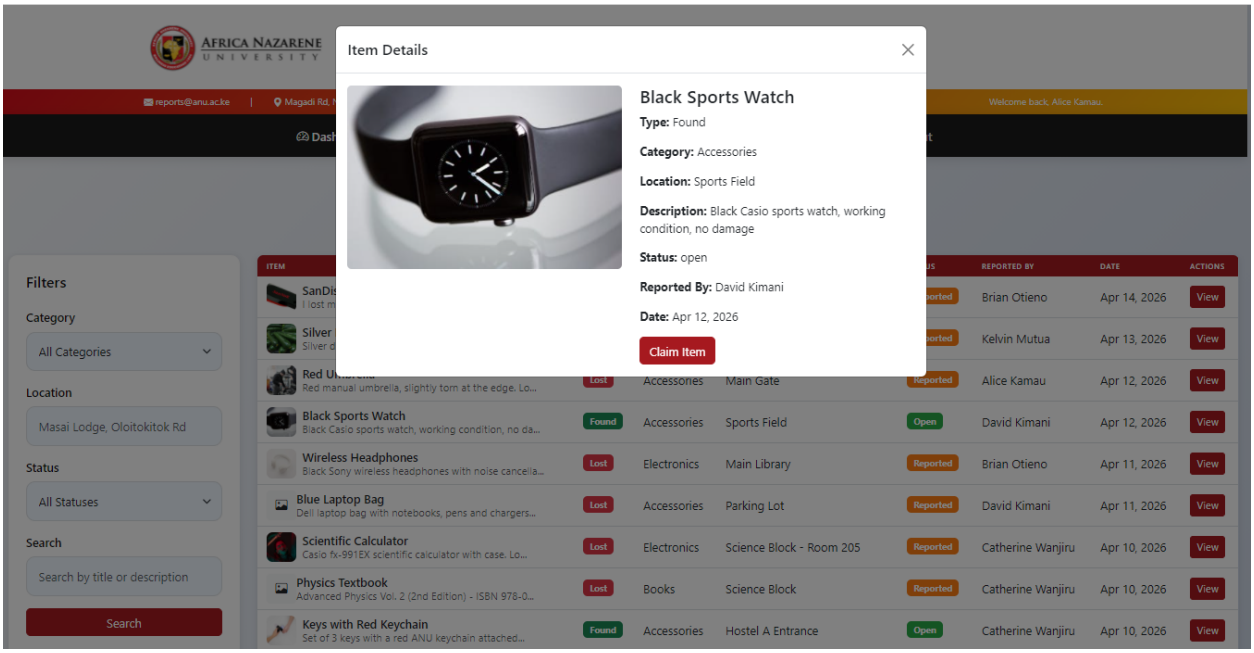

Figure 23: Auto-Match Results Interface

Description:

Shows matched lost and found items generated by the automatic matching algorithm. Students can confirm or deny the match.

---

6. Security Officer Dashboard

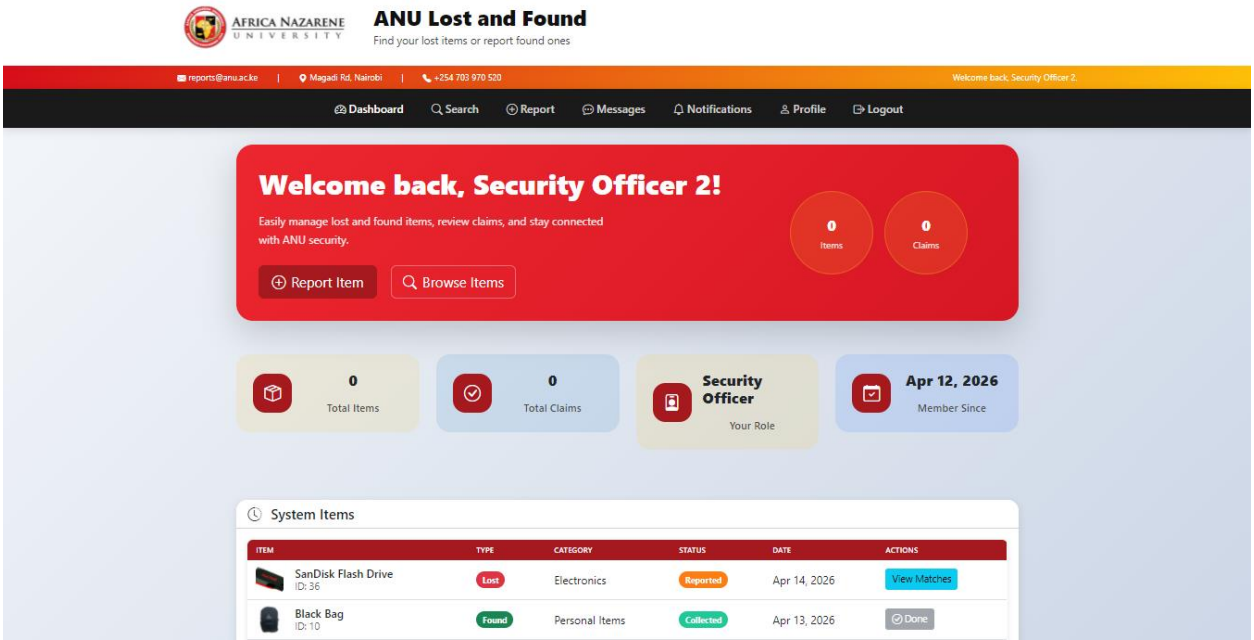

Figure 24: Security Officer Dashboard Interface

Description:

Enables security officers to view pending verifications, verified items, and their profile.

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8. Notifications Screen

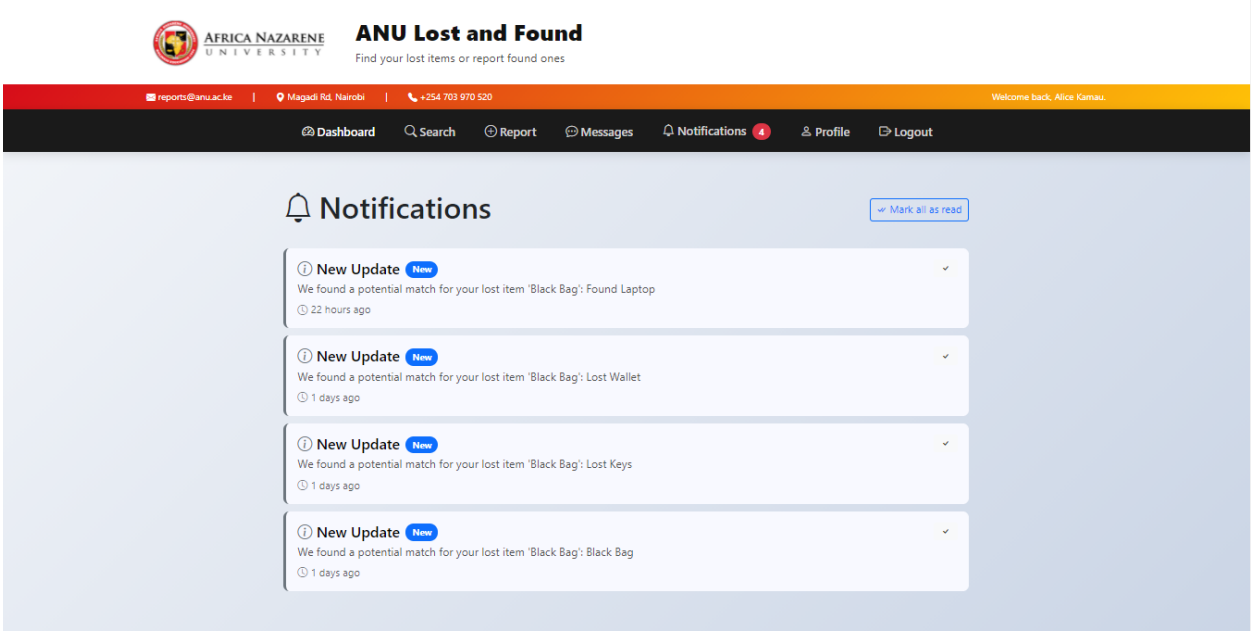

Figure 25: Notifications Interface

Description:

Users receive system alerts such as match results, verification updates, or recovery confirmations.

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9. Administrator Dashboard

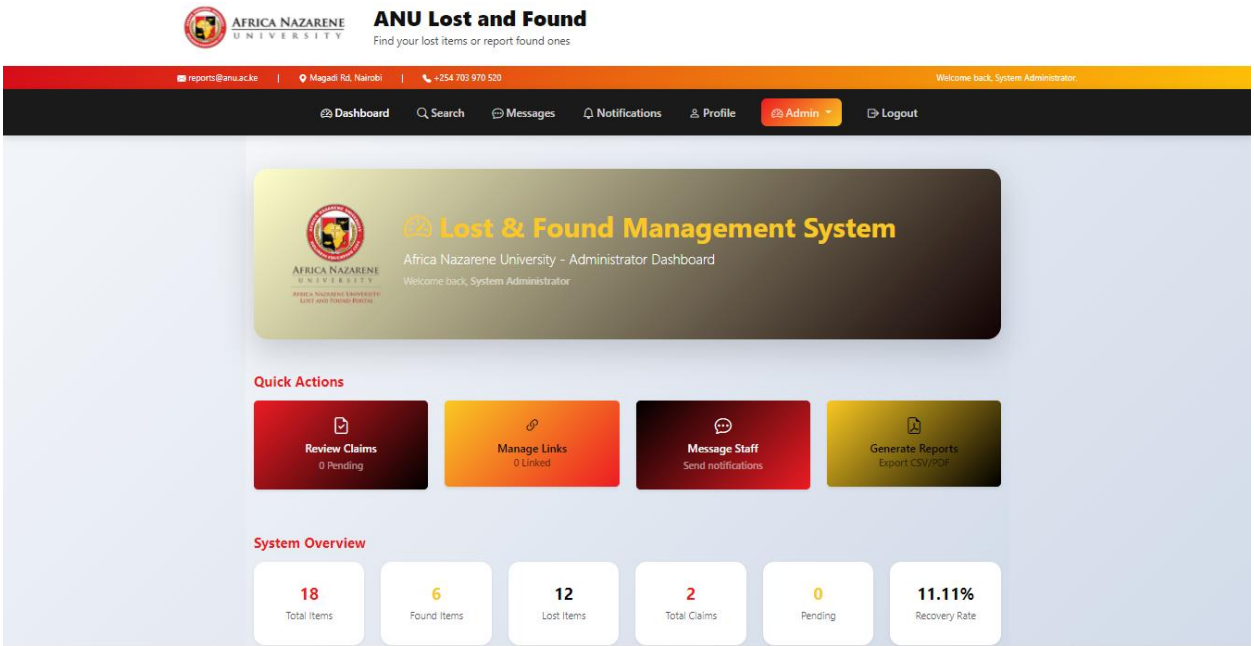

Figure 26: Admin Dashboard Interface

Description:

Used by administrators to generate reports, manage users, manage items, and view system metrics.

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10. Generate Report Screen

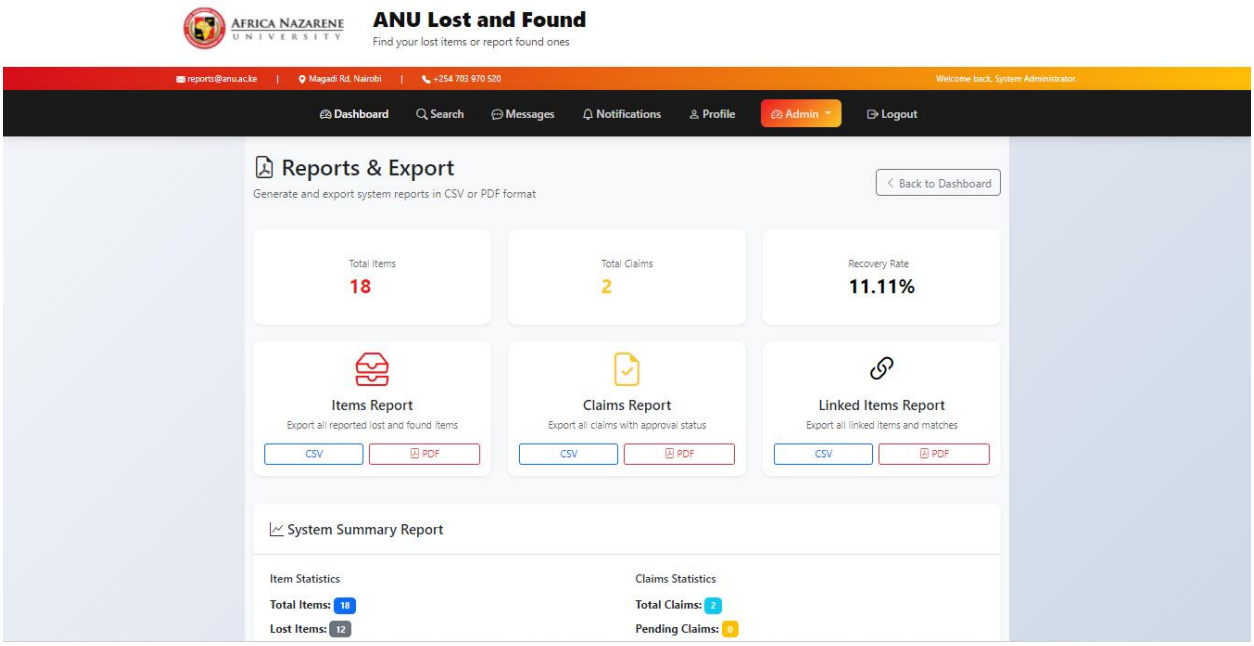

Figure 27: Report Generation Interface

Description:

Admins enter date ranges and generate summaries for lost, found, matched, or recovered items.

A.2 Summary of Prototype Features

The prototype demonstrates the following key functionalities:

✓ Lost & Found Submission

Students and security officers can report lost or found items using structured, guided forms.

✓ Automated Matching

The system automatically compares item descriptions, categories, and metadata to generate match suggestions.

✓ Verification Workflow

Security officers review and validate matches before notifying users.

✓ Notifications

Real-time in-app alerts inform users about new matches, verifications, and final recovery status.

✓ Administrative Reports

Administrators generate monthly and cumulative reports of campus lost and found statistics.

✓ Role-Based Access Control

Each user role has unique dashboards and capabilities based on system permissions.

Appendix B: Questionnaire || Sample Interview Questions

B. 1: Questionnaire for Requirements Gathering

This questionnaire was administered to students, security officers, and system administrators at Africa Nazarene University to collect requirements for the development of the Web-Based Lost and Found Management System. The responses helped identify user needs, challenges with the current manual process, and expectations for the automated system.

Each respondent was informed that their participation was voluntary and that the data collected would be used strictly for academic purposes.

Section A: Demographic Information

1. **Role in the University:**
    - ☐ Student
    - ☐ Security Officer
    - ☐ Administrator
2. **School/Department:**

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3. **Years at ANU:**
    - ☐ Less than 1 year
    - ☐ 1–2 years
    - ☐ 3–4 years
    - ☐ More than 4 years

Section B: Current Lost & Found Experience

4. Have you ever lost an item on campus?
    - ☐ Yes
    - ☐ No
5. If yes, how difficult was the recovery process?
    - ☐ Very easy
    - ☐ Easy
    - ☐ Difficult
    - ☐ Very difficult
6. Where do you currently report a **lost** item?
    - ☐ Security office
    - ☐ Fellow students
    - ☐ Social media groups
    - ☐ Never reported
    - ☐ Other: _____
7. Where do you report a **found** item?

    ☐ Security office

    ☐ Fellow students

    ☐ Social media groups

    ☐ Never reported

    ☐ Other: _____

Section C: Need for a Digital Solution

8. Would a web lost & found application improve the reporting and recovery process?
    - ☐ Yes
    - ☐ No
    - ☐ Maybe
9. Which features would you consider most important? (Tick all that apply)
    - ☐ Report lost items
    - ☐ Report found items
    - ☐ Upload photos
    - ☐ Receive notifications
    - ☐ Automated matching
    - ☐ Verification by security
    - ☐ Track recovery status
    - ☐ Generate reports (Admin only)
10. How urgently do you believe ANU needs this system?
    - ☐ Very urgent
    - ☐ Urgent
    - ☐ Not urgent

Section D: User Expectations

11. What challenges do you face with the current manual reporting system?

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12. What improvements would you want to see in a Web Based Lost & Found system?

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13. Any additional comments?

B. 2: Interview Questions for Requirements Gathering

These interview questions were used during the requirements gathering phase to obtain detailed insights from students, security officers, and administrators about the challenges they face with the current lost and found process, and their expectations for the proposed web-based system.

Section A: General Questions

1. Can you describe your experience with the current lost and found process at ANU?
2. How often do you encounter lost or found items on campus?
3. What challenges do you face when reporting or handling these items?

Section B: System Expectations

4. What features do you believe would be necessary in a Lost & Found web application?
5. How important is real-time notification for lost/found item updates?

6. Would you find automated matching helpful? Why or why not?

Section C: Role-Specific Questions

For Students:

7. Have you ever lost something on campus? What was your experience recovering it?
8. Would you feel comfortable reporting lost items through a web-based app?
9. What information would you expect to fill when reporting an item?

For Security Officers:

10. How do you currently verify found items?
11. What improvements would help make verification easier and more accurate?
12. How often do disputes or false claims occur?

For Administrators:

13. What types of reports do you need for decision-making?
14. How often do you review lost/found statistics?
15. Would automated monthly reports be useful?

Section D: Closing Questions

16. What risks or concerns would you have about implementing such a system?
17. In your opinion, how can this system be improved to serve all users better?
18. Any final suggestions or expectations?